

Collection of problems for Lectures 1-5

Below are a collection of 85 problems related to the material Lectures 1 through 5. At the end of Lectures 1, 2, 3 and 4, I will assign a few questions from this set as the group assignment. Answers to the each assignment are due before the start of the next class. We will post solutions for all of these problems as the week progresses. There will be no assignment for the material in Lecture 5, but please study the questions and suggested solutions when preparing for the midterm exam.

These are group assignments. Each group should submit one solution set for each assignment. The name of every group member who participated in the exercise should be printed at the top of the assignment. One person from the group must attest that the names of the participating members are accurate. If possible, that person should sign the document. You may hand write (rather than type) your solutions – whichever is more convenient. Then scan/photograph your solutions and upload them to the canvas course website.

Problem 1 True, False, Uncertain? [Explain your answer.] Suppose that coffee and cream are demand complements and are traded in competitive markets. Suppose also that there was a reduction in the supply of cream due to a reduction in dairy livestock from disease. *The equilibrium price of coffee will decrease.*

Problem 2 The following demand and supply functions describe the market for natural gas. The g subscripts denote “natural gas” and the o subscripts denote oil. The s and d superscripts refer to “supply” and “demand”, respectively.

$$Q_g^s = 14 + 2p_g + 0.25p_o,$$

$$Q_g^d = 3.75p_o - 5p_g.$$

- (a). Is oil a demand substitute or demand complement to natural gas? How do you know?
- (b). Calculate the equilibrium price of natural gas if the price of oil is fixed at $p_o = 8$.
- (c). Calculate the (point) elasticity of supply for natural gas at the equilibrium price above in (b).
- (d). Suppose for exogenous reasons that the price of oil rises from $p_o = 8$ to $p_o = 16$. What happens to the equilibrium price of natural gas?

Problem 3 The government is interested in the domestic market for wheat. The government’s chief economist estimates the following equations for the demand and supply functions:

$$Q^d = 1500 - 100p,$$

$$Q^s = 500 + 300p.$$

Quantities are measured in thousands of bushels; prices measured in dollars per bushel.

- (a). Calculate the equilibrium price and quantity.
- (b). Calculate the (point) elasticities of supply and demand at the equilibrium values in part (a).
- (c). The government is proposing a \$3/bushel price-support program. What this means is that the government will buy up wheat as necessary to guarantee that the market price will be \$ 3 per bushel. How much wheat will the government need to purchase under this program?

Problem 4 Suppose that there are two kinds of firms that sell plumbing services to a market. The firms are price takers; there is a single market price for an hour of plumbing service. Firms can decide how much to supply at that price. The first type of firm (type “A”) can sell up to 1000 hours of service per week at an incremental (i.e., marginal) cost of \$20 per hour for each hour supplied (this \$20 represents all of the firms additional costs – e.g., the cost of insurance, wages, etc.); beyond 1000 hours of service, it can sell an additional 400 hours of service per week with each additional hour costing \$30 (the higher cost represents the fact that the firm will have to pay overtime wages to its workers). It cannot supply more than 1400 hours per week. The second type of firm, type “B”, is similar to type “A”, except that it has different costs and capacities. It can supply up to 500 hours of service per week at a cost of \$25 per hour; it can supply an additional 200 hours per week for a cost of \$35 per hour; it cannot supply more than 700 hours per week.

Hint: The unit costs given above should be understood to represent each firm’s additional cost of supplying that unit. As such, these costs provide the critical market price at which the firm would be willing to supply service. For example, at a market price of $p = 21$, type “A” firms would supply 1000 and make a “profit” or “producer surplus” of $(21 - 20)(1000) = 1000$ for that supply; type “B” firms would be unwilling to supply at this price since it exceeds its costs of supplying.

Suppose that there are 10 firms of each type in this market (a total of 20 firms). Draw the *market* supply curve. Now suppose that the market price of plumbing services is $p^* = \$32/\text{hour}$. What is the total producer surplus in this market at this price?

Problem 5 Suppose there are 300 buyers and 300 sellers in a market for apples. Each seller has exactly one bushel of apples available to sell; each buyer has demand for at most one bushel of apples to buy.

Suppose 200 sellers have a cost of \$10 per bushel and 100 sellers have a cost of \$30 per bushel. These costs represent what the sellers give up by selling a bushel of apples (i.e., this is their opportunity cost of the apples). Suppose 200 of the buyers have a value for apples of \$20 per bushel and 100 buyers have a value of \$40 per bushel. These values represent the most that the buyers are willing to pay.

- (a). What is the equilibrium price in this market and how many bushels of apples are traded? Determine this outcome by drawing the associated supply and demand graphs.
- (b). How much “profit” do the sellers collectively make? (Hint: This is the area between the market price in (a) and the supply curve.)
- (c). How much “consumer surplus” do the buyers obtain? (Hint: This is the area between the demand curve and the market price in (a)).
- (d). Suppose the government taxes the sellers \$5 per bushel sold. What is the new equilibrium (pre-tax) price? (Hint: the tax is equivalent to raising the seller’s cost of a bushel of apples by exactly \$5. You can repeat the analysis in (a) with this modification.) What happens to producer surplus and consumer surplus? Would it matter if the government taxed the buyers instead (i.e., each buyer’s willingness to pay decreased by \$5)?

Problem 6 In an effort to reduce carbon emissions, the US government has undertaken a broad program of incentives and regulation to eliminate all oil consumption. It anticipates the program will reduce oil consumption from 20 million barrels of oil per day to zero barrels per day, and it anticipates doing this without any substitution into carbon-emitting energy sources.

Suppose that the US estimates are correct and that the elimination of all oil consumption by the US will be accomplished. Does this imply that the US policy will reduce the world’s consumption of oil by 20 million barrels of oil per day? Explain what determines the amount of the world reduction.

Problem 7 An extremely wealthy environmentalist has decided that he wants save rain forests from lumber mills and deforestation. To this end, he buys 100,000 acres of dense rain forest. The agreement gives him exclusive rights to the resources of the forest for this time period. Normally, a purchaser of such a lease would strip the land of its lumber. Rather than logging the forest for valuable mahogany, however, the billionaire chooses to make the forests into nature preserves.

Given his forest purchase includes approximately 1 million mahogany trees (or 100 million cubic feet of wood), he concludes that he has reduced the world’s consumption of rainforest by 100,000 acres and rain-forest mahogany by 100 million cubic feet.

Has the billionaire actually reduced the world’s consumption of rain forests and mahogany by these amounts? What does the conclusion depend upon?

I apologize for the confusing solution to this question. Let me try again here.

There are two ways to solve the problem, each is equally valid.

Problem 8 The following demand and supply curves describe the market for wheat. The w subscript denotes “wheat,” the sb subscript denotes “soybeans” and the c subscript denotes

“corn”. The s and d superscripts denote “supply” and “demand”, respectively.

$$Q_w^s = 40 + p_w,$$

$$Q_w^d = 4p_{sb} + 7p_c - 10p_w.$$

- (a). Are soybeans a substitute or a complement for wheat? How do you know?
- (b). Without any mathematical calculations, describe how the demand curve for wheat shifts as the price of corn rises and everything else remains unchanged.
- (c). What is the equilibrium price of wheat when $p_{sb} = 50$ and $p_c = 40$? What is the equilibrium quantity of wheat traded?
- (d). At the prices above in (c), what is the point elasticity of demand for wheat with respect to the price of wheat?

Problem 9 If the supply and demand curves are given by $Q_d = 30 - p$ and $Q_s = 5 + 4p$, compute the point demand and supply elasticities at equilibrium.

Problem 10 True, False, or Uncertain. [Explain your answer!] *If the demand for Intel-based personal computers is inelastic, then the demand for the Gateway brand of computers (Gateway is a specific brand of Intel-based personal computers) is also generally inelastic.*

Problem 11 Suppose that demand for beer is given by

$$Q_{beer}^d = 120 - 4p_{beer} + 3p_{wine} + 2I,$$

and supply is given by

$$Q_{beer}^s = 100 + 6p_{beer}.$$

- (a). Suppose that $p_{wine} = 20$ and $I = 10$. What is the equilibrium price and quantity of beer?
- (b). Income elasticity:
 - (i). What is the income (point) elasticity of beer at this equilibrium price?
 - (ii). Is beer an inferior good or a normal?
 - (iii). Is it a luxury good?

Problem 12 Consider a market in which the supply curve for memory chips has an elasticity of $\varepsilon_s = 2.0$ and the demand curve has an elasticity of $\varepsilon_d = -0.5$.

- (a). Following an isolated earthquake, the production facilities of several manufacturers are destroyed; no negative effects, however, occur on the demand side of this market. If total output sold falls 10%, approximately how much did the price rise (in percentage terms)?

Explain your answer. [You may assume the given elasticities are constant over this change in output.]

(b). Does the earthquake described in (a) increase or decrease market revenues? Explain.

(c). Suppose that over time, the damaged production facilities are repaired and supply shifts back to precisely its original position. Unfortunately, demand for chips decreases due to reduced demand for computer equipment. Does revenue increase or decrease relative to the original (pre-earthquake) equilibrium? Explain.

Problem 13 True, False, Uncertain? [Explain your answer.] Currently, the supply and demand curves for cigarettes intersect at a price of \$2. Suppose that the elasticity of demand is $\varepsilon = -2.0$. *If the quantity demanded strictly decreases at every price due to a change in tastes, then the equilibrium price will decrease and revenue will increase.*

Problem 14 True, False, Uncertain? [Explain your answer.] Suppose the demand elasticity for oranges is elastic; specifically, suppose that $\varepsilon_d = -2$. *When the market supply for oranges decreases due to bad weather, consumer surplus in the market decreases.*

Problem 15 If the demand elasticity is larger than 1 in absolute value and the supply curve shifts right, do total market expenditures increase or decrease? Explain.

Problem 16 True, False, Uncertain? [Explain your answer.] The July 20, 1999 edition of the *Wall Street Journal* states:

Philip Morris Inc., the company's U.S. tobacco unit, reported operating income of \$1.2 billion, down 3.9% from last year, with lower volume offset by higher prices. The unit's revenue rose to \$4.74 billion from \$3.73 billion, but total volume fell to 51.8 billion cigarettes shipped, down 9.6% from last year.

Therefore, the demand for Cigarettes is elastic.

Problem 17 The demand for haircuts is

$$Q^d = 10,000 - 200p,$$

while the supply of haircuts is

$$Q^s = 1200p - 18,000,$$

assuming that $p \geq 15$; $Q^s = 0$ if $p < 15$.

(a). What is the equilibrium price of haircuts in this market?

(b). What is the own-price elasticity of demand in equilibrium? Is demand elastic or inelastic?

- (c). Compute the consumer and producer surpluses at the equilibrium price in (a).
- (d). Now suppose that supply decreases slightly to

$$Q^s = 800p - 12,000,$$

if $p \geq 15$ and $Q^s = 0$ if $p < 15$. The demand relationship remains unchanged. What is the new equilibrium price? Is demand elastic or inelastic?

- (e). Are market revenues larger at the new price in (d) or at the original price in (a)? Could you have answered this question by using your observations about the elasticities of demand? Explain.

Problem 18 Draw indifference curves reflecting the following statements:

- (a). Advertising statements to the contrary, Coca-Cola and Pepsi-Cola are all the same to me. I don't care which one I get.
- (b). I never eat a Big Mac without a Coca-Cola and I never drink a Coca-Cola without a Big Mac.
- (c). I like apples and oranges. But if you take away half my oranges and double the number of apples I have (and vice versa), I'm just as well off.

Be sure your graphs are well labeled and clearly explained.

Problem 19 True, False, or Uncertain. [Explain your answer!] Suppose a consumer's utility curve is given by $U(x, y) = 2x + 3y + 7$. *Here the goods x and y exhibit diminishing marginal rates of substitution.*

Problem 20 A consumer has income of \$200. Good x costs \$4 per unit and good y costs \$10 per unit.

- (a). Graph the consumer's budget constraint.
- (b). Show what happens to the budget constraint if the price of good y falls to \$5.

Problem 21 You are managing a city budget in which monies are spent on law enforcement and other services (e.g., schools, etc.). Suppose that the city has 5 million Euros of its own funds to allocate between law enforcement and all other services.

You are informed that you are about to receive aid from the federal government to support a special anti-drug law enforcement program. Two programs which might be available are (1) a 1,000,000 Euro grant which must be spent on law enforcement; (2) a matching grant program, in which each euro of local spending on law enforcement is matched by a euro of

federal money which must be spent on law enforcement. The federal matching program limits its payment to each city to a maximum of 1,000,000 Euros.

- (a) Draw the budget constraint relevant for each program.
- (b) Which program would you (the manager) choose if you wish to maximize the satisfaction of the citizens in your jurisdiction?

Problem 22 True, False, Uncertain? [Explain your answer!] If your marginal utility of pizza is larger than your marginal utility of beer, *you should purchase more pizza and less beer.*

Problem 23 True, False, or Uncertain. [Explain your answer!] If Al's income and the prices of every good in the economy simultaneously increase by 10 percent, with the sole exception of the price of beer which remains unchanged, *then Al's consumption of beer must increase.*

Problem 24 True, False, or Uncertain? [Explain your answer!] Robert's marginal utility from consuming chocolate for dessert is 7 and his marginal utility from consuming strawberries for dessert is 4. The price of a dessert of strawberries is £2 and the price of a dessert of chocolate is £3. *Robert should consume chocolate for dessert.* [You may assume that Robert can only consume one or the other for dessert.]

Problem 25 True, False, Uncertain? [Explain your answer.] Suppose that a consumer's indifference curves are convex and that the consumer has optimally allocated his food budget. At the present levels of consumption, the consumer's marginal utility of consuming "pasta" is 9 utils/package and the marginal utility of consuming "steak" is 15 utils/lb. *If the price of steak is currently 5 dollars per pound, then the price of pasta must be 2.50 per package.*

Problem 26 Suppose that Diane's marginal rate of substitution between wine and beer always equals

$$MRS_{\text{wine,beer}} = 3;$$

that is, she would give up three bottles of beer for one bottle of wine at any consumption bundle of beer and wine.

- (a). What can we say about the relationship between these two goods?
- (b). If the price of beer is $p_b = 2$ and the price of wine is $p_w = 7$, what ratio of wine and beer should Diane consume?

Problem 27 (10 points) True, False, or Uncertain. [Explain your answer!] In a two-good world (goods x and y), if the prices of good x and y increase by 5 percent and the income of the consumer decreases by 5 percent, *the marginal rate of substitution of the customer between goods x and y will increase at the optimum.*

Problem 28 True, False, or Uncertain. [Explain your answer!] A consumer's consumption consists of only two goods, good A and good B. When the price of good A rises, the consumer's total expenditures on good B rises. *The consumer's demand for good A is price inelastic.*

Problem 29 True, False or Uncertain? [Explain] The price of Australian wine is \$10 per bottle and the price of French wine is \$25 per bottle. Dominique's marginal utility from her last bottle of Australian wine is 120 utils, while her marginal utility from the last bottle of French wine is 300 utils. *She has rationally allocated her budget across Australian and French wine.*

Problem 30 Consider a simple setting with two goods, x and y . The price of x is $p_x = 20$ and the price of y is $p_y = 15$. Due to various government regulations, consumption is restricted to no more than 2 units of either good (i.e, your consumption choice must satisfy $x \leq 2$ and $y \leq 2$). Consumption is divisible (e.g., it is possible to consumer $x = 1.62$ units). Your income is $I = 60$.

(a). Graph your constraints for x and y in this context. That is, indicate the set of consumption choices that are feasible (affordable and satisfy the government regulations).

(b). Suppose that your marginal utility of x is 240 and your marginal utility for y is 200 utils, regardless of the amounts you consume of either good. Exactly how much x and y should you optimally consume given your constraints from part (a).

Problem 31 True, False, Uncertain? [Explain your answer.] *In a two-good world in which both goods are demand substitutes, both goods must have elastic demand curves.*

Problem 32 True, False, Uncertain? [Explain your answer.] Suppose that when the consumer's income rises by 20%, the consumer's consumption of good x only increases 10%. *We can infer that good x is an inferior good.*

Problem 33 A town wants to repair its swimming pool. The annual demand from each of its 5,000 citizens in terms of uses is $q = 18 - 3p$ where p is the price in (dollars) charged for using the pool. If it charges a \$1 admission, it will raise \$75,000 a year, enough to pay for the repairs. An alternative proposal is to charge \$15 a year for a season ticket with no per-usage fees. Which proposal would be better from the point of view of the consumer? Can you quantify the relative advantage of the superior proposal?

Problem 34 When planning this year's budget, Helen Smith decided to purchase 40 compact disks this year at 5 pounds each and to spend the remaining amount of her income on other goods. Before purchasing anything, however, she is offered an opportunity to buy her compact disks from a music club, where she must pay 6 pounds a disk for the first 20 disks but only 4 pounds per disk thereafter. Is she indifferent to the offer, or should she take it? Explain your answer.

Problem 35 The price of apples increases 10 percent, while the price of all other goods (AOG) remains constant. You respond by reducing your consumption of apples by 5 percent.

- (a). Is your demand for apples elastic or inelastic? Explain.
- (b). What happens to total expenditures on apples? Explain.
- (c). Does your consumption on all other goods beside apples increase or decrease? Explain.

Problem 36 Chicago Corp. has its headquarters in Chicago and a branch office in San Francisco. Recently, whenever Chicago Corp has transferred employees to its San Francisco office, it has raised salaries by 30 percent to take into account the higher cost of housing in the San Francisco Bay area. [You should assume that other goods are roughly the same price as in Chicago; only housing is more expensive.] Nonetheless, while turnover in the Chicago office is only 10 percent per year, the turnover in the San Francisco branch by recently transferred employees is 40 percent. The common complaint by quitting employees is that the cost of living adjustment was insufficient to maintain their standard of living, and hence they preferred to take a job at another company in the Bay area after having settled in.

Taking the transferred workers' complaints seriously, you decide to look carefully at housing prices in the Bay area and adjust their new salaries so as to allow them to purchase the same bundle of goods as if they had remained in Chicago. Your review of the relative housing prices and your resulting calculations suggest that you need to raise their salaries by a full 50 percent rather than merely 30 percent. You are convinced that with this salary adjustment, the transferred workers will be able to purchase the same bundle of goods (housing and other stuff) as they consumed in Chicago.

One year later when you evaluate the annual turnover rates, you notice that while the Chicago turnover rate remains at 10 percent per year, the San Francisco turnover rate has declined to 1 percent implying that you have paid them too much! What was wrong with your cost-of-living-adjustment (COLA) analysis? Draw a graph to illustrate what you have done and what has gone wrong.

Problem 37 (a). Consider a random variable that takes on three possible values, $x_1 = -10$, $x_2 = 15$, $x_3 = 25$, with probabilities $\phi_1 = \frac{1}{2}$ and $\phi_2 = \phi_3 = \frac{1}{4}$. What is the expected value of the random variable?

(b). Consider a second random variable which also takes on three values, $y_1 = -25$, $y_2 = 25$, $y_3 = 45$ with probabilities $\phi_1 = \frac{1}{2}$ and $\phi_2 = \phi_3 = \frac{1}{4}$. What is the expected value of this second random variable?

(c). If a risk averse person were to choose between these two assets, which one would be chosen? [You may assume that the risk averse person has mean-variance preferences.]

Problem 38 Suppose that an Faruk has wealth of \$100,000 and is presented with the following project: with probability $\frac{1}{4}$, the project succeeds and generates profit of \$10,000; with

probability $\frac{1}{2}$, profits are zero; with probability $\frac{1}{4}$, losses are \$10,000. Suppose that the money utility function, $u(x)$, satisfies the following: $u(110,000) = 30$ utils, $u(100,000) = 26$ utils, $u(99,000) = 25.5$ utils and $u(90,000) = 20$ utils.

- (a). What is the expected value (not expected utility) of the project?
- (b). Should Faruk take the project or not?
- (c). If Faruk takes on the project, what is his risk premium?

Problem 39 Suppose that two possible events can occur with a particular project: the Good outcome and the Bad outcome. If the Good outcome occurs, your company *receives* \$400. If the Bad outcome occurs, your company *loses* \$200. Suppose that you (and your company) are risk neutral.

- (a) If the probability of the Good outcome is $\phi_G = .5$, what is the expected value of this project?
- (b) Suppose that you do not know the probability of the Good outcome. What value of ϕ_+G would make you indifferent to undertaking this project?
- (c) Now suppose that your company has limited liability for debts equal to your current assets which are \$100. This means that if you take on this project, your downside losses are limited to \$100 because that is all that you must legally pay if the Bad event occurs. Given your limited liability, what is the probability of the Good outcome that would make you indifferent to undertaking this project? That is, answer (b) again, but for the case in which downside losses are limited to \$100.
- (d) At the probability you found in (c), what is the expected value of this project if you had instead \$200 in assets (i.e., you could cover the entire loss of \$ if the Bad outcome arises)?
- (e) Comparing (b) and (c), it appears that for some probability of the good outcome, a risk-neutral company with sufficient assets would reject the project but a risk-neutral company with limited assets and limited liability would accept. It is as if the company is risk loving. Explain why this happened using concepts from class.

Problem 40 True, False, Uncertain? [Explain your answer.] Every additional dollar Joan receives as income provides less additional utility than the previous dollar. *Joan is therefore risk loving.*

Problem 41 For simplicity, suppose there are two kinds of used cars which exist in a 50-50 proportion: good cars worth 5,000,000 pesetas and bad cars (lemons) worth only 1,000,000 pesetas. Also suppose that only the owner truly knows the type of car that it is.

- (a) If both kinds of cars are sold, what is your expected value from buying a used car?

(b) Now suppose that the current owners of good cars (i.e., the sellers) have a reservation value of 4,000,000 pesetas (i.e., they'd rather keep a good quality car than sell it for less than 4,000,000 pesetas); the sellers of bad cars value their cars at only 800,000 pesetas. You know that these reservation prices exist, you just can't distinguish the good car sellers from the bad car sellers. If someone offers you a car at a low price, say 2,500,000, do you buy it? Why or why not?

(c) Is there any market price at which uninformed buyers (i.e., buyers who know everything stated above *but* who do not know whether the car is good or bad by just looking at it) will buy used cars and informed sellers will sell them? What type of cars are sold?

(d) What general problem is this related to?

Problem 42 Elaine has an idea for a small business which is risky. With probability of 0.5, the business will succeed, resulting in profits of \$5,000; with probability of 0.5, the project will be a failure and produce losses of \$2,500. Elaine's initial wealth is \$5,000. The time horizon is sufficiently short that discounting can be ignored.

(a) What is the expected value of the project?

(b) Elaine's utility function for income is approximately $u(x) = \sqrt{x}$. Is she risk averse? Is her marginal utility of income increasing or decreasing?

(c) What is Elaine's expected utility if she undertakes this small business venture? [Hint: Remember that expected utility is calculated by considering the different resulting *wealth* levels which may arise from undertaking a project.] What is Elaine's expected utility if she does nothing and maintains her \$5,000 wealth position with certainty? Should she undertake the project?

(d) What risk premium would Elaine be willing to pay in exchange for receiving the expected value of the project for sure rather than the uncertain gamble?

(e) Suppose that her friend, Kramer, is also thinking about starting a business with identical prospects for success and failure (and identical profits and losses), but the proposed business is completely unrelated to (i.e., independent of) Elaine's project. Elaine and Kramer decide to form a partnership, agreeing to split evenly any gains and/or losses between themselves. Does Elaine's expected utility increase, decrease, or remain the same under the partnership arrangement? Briefly explain.

(f) If Kramer and Elaine weren't such good friends and Elaine couldn't observe how hard Kramer was working, what general problem should Elaine be concerned with? Briefly explain.

(g) Thinking that their partnership is a good idea, Elaine and Kramer decide to place an ad in the paper to get other entrepreneurs with ideas to join their partnership, thereby reducing the risk of their business portfolio. What problem should Elaine and Kramer be worried about, in addition to the problem above in (f)? Briefly explain.

Problem 43 True, False, or Uncertain. [Explain your answer!] The following gamble is offered to a risk-averse person free of charge: win £2000 with probability $1/2$, win £1000 with probability $1/4$, lose £5000 with probability $1/4$. *A risk averse person will not accept this gamble.*

Problem 44 Ellsworth faces a probability of 0.5 that his boss, Gail, might fire him. If he is fired, Ellsworth will remain unemployed for a year, earning nothing. His yearly salary is \$360,000.

(a) If he prefers earning \$180,000 with a guarantee of employment to his current situation of receiving \$360,000, then what can we say about Ellsworth's preferences toward risk?

(b) Now suppose that he would be willing to take a pay cut of up to \$200,000 per year (from his current income of \$360,000) in exchange for a guarantee of not being fired; what is his risk premium?

(c). What might be a reason why employers, like Gail, would choose *not* to offer an employment guarantee such as in (b)?

Problem 45 Suppose that a consumer has the following demand curve:

$$p = 10 - 2q.$$

Because of uncertainty over supply, the market price can be either $p = 6$ (normal supply) or $p = 8$ (shortage of supply). These events occur with equal probability (i.e., $\phi_1 = \phi_2 = \frac{1}{2}$).

(a). What are the consumer surpluses in each of the two supply states.

(b). Suppose now that the consumer has to spend \$2 in order to access this market, but that the money must be spent *before* the consumer knows the supply situation. (For example, the consumer must take a day-long journey to get to the market place, and the journey costs him \$2.) What is the expected monetary value of traveling to the market (remember to subtract the costs of travel). Will a risk-neutral consumer wish to travel to this market?

(c). Now suppose our consumer has preferences over money given by $u(x) = x^2$ and the consumer's wealth is at least 14 (i.e., they can afford the \$2 expense to access the market and have enough money leftover to purchase according to the inverse demand curve $p = 10 - 2q$). Assume that consumer surplus is equivalent to money for this consumer. Will the consumer want to travel to the marketplace? (Note: you may want to plot a few points of $u(x) = x^2$ to understand this person's preferences for risk. Also, you may be able to answer this question without calculating the expected utility of the market trip.)

Problem 46 You manage the marketing department for Vanboard Brokerage Services. The accounting department has advised you that, on average, every new individual account generates an economic profit of \$100 per year, and that the average new customer stays with

Vanboard Brokerage Services for more than 2 years. Your own marketing surveys indicate that if you offer a promotion of a gift certificate worth \$50 to anyone who starts a new customer account at VBS, that you will attract 1000 new customers. What idea from uncertainty theory suggests that this may not be a profitable promotion? Explain.

Problem 47 Suppose that an investor's utility for money is $u(x) = \sqrt{x}$, her wealth is \$1,000,000, and that she is offered an investment opportunity which yields a *loss* of \$990,000 with probability $\frac{1}{2}$ and a *gain* of \$2,240,000 with probability $\frac{1}{2}$. Should she take the gamble? Explain. [Hint: $\sqrt{3,240,000} = 1800$.]

Problem 48 At some international airports, there are multiple passport control agents, with a single line serving several agents. (E.g., Heathrow airport.) When it is your turn, in such a one-line system, you go to the next available agent. At other international airports, each passport control agent has a separate line, and it is possible that if you choose badly, you will take longer to get through passport control than the average traveler; of course, if you choose well, you may end up taking much less time than average in such a multi-line system. (E.g., Frankfurt and Singapore airports.)

Suppose that everyone prefers to get through passport control as quickly as possible, and that no one has a superior advantage at guessing which lines will move faster. If the utility of a person's time can be represented in a similar manner as the utility of a person's money, explain why one system may be preferred to another depending upon a traveler's marginal utility of time. [Hint: you may want to think about risk preferences and diversification.]

Problem 49 True, False, Uncertain? [Explain your answer.] In the textbook model of costs, suppose that at the current level of output, the long-run average cost curve is decreasing, but the short-run average cost curve is increasing in output. *It follows that the long-run marginal cost curve must be below the short-run marginal cost curve at this level of output.*

Problem 50 A firm's cost function is $C(q) = q^2 + 16$. What is its marginal and average cost functions? Where is the minimum of average cost?

Problem 51 True, False or Uncertain? [Explain your answer.] A firm's long-run costs are currently \$100, of which \$50 represents factor payments to workers and \$50 represents payments to capital. *If wage rates increase by 50%, total long-run costs will go from \$100 to \$125.*

Problem 52 Suppose that a competitive firm can hire as many workers as it wants at the going wage of \$150 per day and can employ robots (capital) at a constant daily cost of \$300 each. At present, the firm is producing 200 cars per day with an input combination for which $MPP_K = 0.3$ and $MPP_L = 0.1$.

(a) What is this firm's marginal rate of technical substitution currently? Is this firm using an economically efficient input combination? If not, what should it do?

(b) In the short run (your robot level is fixed, but workers are variable), what are your marginal costs for an additional car at the present level of production?

Problem 53 True, False, or Uncertain? [Explain your answer!] *The long-run opportunity cost of producing a given level of output is always less than the short-run opportunity cost of producing that level of output.*

Problem 54 True, False, or Uncertain? [Explain your answer!] *If wages increase, then the short-run marginal cost of a competitive firm necessarily increases, while the long-run marginal cost could increase or decrease depending upon the particulars of the situation.*

Problem 55 True, False, or Uncertain. [Explain your answer!] All input prices rise by exactly 10 percent. *Both long-run and short-run marginal costs will increase by exactly 10 percent.*

Problem 56 True, False, Uncertain? [Explain your answer.] If average cost is constant for all levels of output, *then the marginal cost curve is flat and equal to the average cost curve.*

Problem 57 True, False, Uncertain? [Explain your answer.] For a given technology, whenever labor inputs double and capital inputs are held constant, output less than doubles. *This technology exhibits decreasing returns to scale.*

Problem 58 True, False, Uncertain? [Explain your answer.] Suppose that the prices of all short-run flexible inputs (e.g., labor) increase by 10 percent and short-run fixed inputs (e.g., capital) increase by 20 percent. *The short-run marginal cost of output will increase by more than 10 percent but less than 20 percent.*

Problem 59 Eric Clapton concert tickets cost \$45. You won a free ticket to see the Eric Clapton concert (which is not transferable and has no resale value). Bob Dylan is performing on the same night and is your next-best alternative activity. Both performers are only performing on this one night. Tickets to see Dylan cost \$40. You would be willing to pay up to \$50 to see Dylan if you don't go to the Clapton concert. Assume there are no other costs of seeing either performer. Based on this information, what is your opportunity cost of seeing Eric Clapton?

Problem 60 True, False, Uncertain? [Explain your answer.] Yesterday you were unexpectedly given a free ticket to an Coldplay concert scheduled for August 1. The market price of the ticket is \$75, but you can only sell it for \$50 if you don't use it yourself. Today, you

discover that U2 will be giving a concert that same evening (and only that evening). Tickets for the U2 concert are available at \$100. Had you not received the free Coldplay ticket, you would have bought a U2 ticket rather than a Coldplay ticket. *From what we are told of your preferences, it follows that if you are a rational utility maximizer, you should attend the U2 concert.*

Problem 61 Suppose that a competitive firm's cost of producing output q in the short-run is given by $C_{sr}(q) = 3q + q^2$. You may assume that there are no short-run capital expenditures, so variable cost and short-run cost are identical. Assume that for this firm all costs are variable in the short-run. The market price of the firm's product is \$9.

(a). What level of output will the firm produce?

(b). What is the firm's profit?

Problem 62 True, False, Uncertain? [Explain your answer.] *At the optimal point of production, if a firm's marginal revenue is above its average (opportunity) cost, then average (opportunity) cost must be increasing.*

Problem 63 In late July, a toy company receives an order for *up to* 75,000 toys to be delivered November 1. The contract calls for a price of \$20 each, payable on November 1. You may fulfill any portion of this order; you do not need to produce all 75,000 toys. Total monthly manufacturing costs, as a function of the number (in thousands) produced that month, are

$$C(q) = 80 + 0.5q^2$$

[You may assume that 80 is sunk if you like; your answer should not change.] Because this is such a large order, the company will be required to stockpile production until the November shipping date. However it will be required to pay storage costs of \$1 per toy per month until November 1. That is, a toy made in August will incur a storage cost of \$2 (September and October). (Worries about finance charges incurred between August 1 and November 1 should be put aside.) What is the optimal number of toys to be produced each month?

Problem 64 True, False or Uncertain [Explain your Answer!] *A firm should always produce at an output at which long-run average cost is minimized.*

Problem 65 You are in charge of a small aspirin manufacturing company. Each day you can make up to 1000 cases of generic aspirin. You can sell as many cases as you make for the market price of \$10 per case. Your weekly fixed operating costs are \$1000 (for rent and insurance). The cost to you for materials and supplies is \$4 per case; the cost of labor is \$5 per case, except on weekends, when it is \$10 per case (because of overtime labor costs). There are no sunk costs in producing aspirin.

(a). How many cases of aspirin should you make in a week? What is your weekly economic profit?

(b). Now suppose that in addition to making generic aspirin, you can also make a special “branded” version with the same costs as the generic version. You receive a special order for 1000 cases of branded aspirin per week for \$30 per case. Note that this is your only demand for your “branded” aspirin; your other production will be generic aspirin. How many cases of branded and generic aspirin should you produce each week?

Problem 66 A firm produces its first 30 units at a cost of \$ 10 per unit. Its next 20 units costs \$15 per unit. All additional units cost \$20 per unit. There are no fixed costs of production, so the marginal cost is equal to the average (opportunity) cost.

The firm can sell up to 40 units at a price of $p_A = \$12$ in market A , and up to 20 units at $p_B = \$17$ in market B . Nothing will be demanded at higher respective prices, and the same amount will be demand in each market at lower respective prices. Finally, assume that the firm’s costs only depend upon the total amount of output produced across both markets and that consumers cannot transport goods across market boundaries.

How much should the firm produce in each market?

Problem 67 Suppose that a firm can sell as much output as it likes at the market price of $p = 20$. Its opportunity cost of production in the short-run $60 + 2q^2$. (This implies that $MC_{sr}(q) = 4q$.) Its opportunity cost of production in the long-run is $90 + q^2$. (This implies that $MC_{lr}(q) = 2q$.)

(a). What is the profit maximizing level of output in the short-run? Should the firm operate or shutdown? Explain.

(b). What is the profit maximizing choice of output in the long-run? Should the firm operate or shutdown? Explain.

Problem 68 True, False, Uncertain? [Explain your answer.] Suppose that at a competitive firm’s current level of output, $p < MC$ and $p > AC$, where all costs are opportunity costs. *This firm could increase profit by increasing output.*

Problem 69 True, False, or Uncertain [Explain your Answer!] *A price low enough to cause a firm to shut down in the long run may not be low enough to cause it to shut down in the short run.*

Problem 70 Suppose that in a market with perfect competition, the market demand function is given by $Q^d = 1200 - 50p$. Every firm in the industry has a short-run *opportunity* cost of production given by

$$C_{sr}(q) = 2q^2.$$

- (a). Compute each firm's short-run supply curve.
- (b). If there are 600 firms, what is the short-run equilibrium price and quantity in this market?
- (c). If each firm's long-run *opportunity* cost function is given by

$$C_{lr}(q) = 2q^2 + 2 + q,$$

and there are an unlimited number of firms available with this cost function, what is the long-run equilibrium price in the market? How many firms will be producing?

Problem 71 The cheese business in Wisconsin is a competitive industry with an unlimited number of potential suppliers. All cheese manufacturers have the long-run opportunity cost function $C_{lr}(q) = q^2 + 9$. The demand for cheese manufactured in this town is given by

$$Q^d = 120 - p.$$

What is the long-run equilibrium number of firms in this industry?

Problem 72 Electricity is supplied competitively by two types of firms. The first type of firm has a very efficient (i.e., low-cost) form of power generation; this component of the market (which consists of dozens of competitive producers) will supply up to 10 Million megawatt hours (MMH) per day at a price of 1 per MMH. At any lower price, this market segment will supply nothing, but due to capacity constraints, a higher price than 1 will not produce greater supply than 10 MMH. [Hint: the supply curve for this segment is horizontal to 10 MMH, and then it becomes vertical.] The second segment of the market is less efficient, but similar in form. In aggregate, it will supply up to 5 MMH at a price of 2 per MMH. At any lower price, the inefficient segment will not supply any electricity; at a higher price than 2, this segment will only supply 5 MMH due to capacity concerns.

- (a). Graph the market's aggregate supply curve carefully labeling the points of interest on the curve. At a price of $p = \frac{1}{2}$, how much is supplied by the market? At a price of $p = \frac{3}{2}$, how much is supplied? At a price of $p = 3$, how much is supplied?
- (b). Suppose that the market demand curve for electricity is given by

$$P(Q) = 16 - Q.$$

What is the equilibrium price in this market? [Hint: graph the supply and demand curves rather than using algebra.] How much does each segment of suppliers produce at this equilibrium?

- (c). Suppose that government regulation forces the closure of all the inefficient plants due to pollution problems. What is the new equilibrium price and quantity?

Problem 73 Three firms are now manufacturing widgets for sale in a highly competitive market. Each firm has a capacity of seven units. Their costs are as follows:

Quantity	Firm A	Firm B	Firm C
1	1	1	1
2	3	4	2
3	6	8	4
4	10	13	7
5	15	19	11
6	21	26	16
7	28	34	22

The market demand for widgets equals: $25 - 2p$

- What is each firm's marginal cost and supply schedules?
- How many are produced? What is the equilibrium price?
- By which firms?

Problem 74 Suppose that in a market with perfect competition, the market demand function is given by $Q^d = 96 - p$ and the market supply function is given by $Q^s = 3p$. Every firm in the industry has a short-run opportunity cost of production (i.e., “variable” cost) of $C_{sr}^O(q) = 4q^2$, which has an associated marginal cost of $MC_{sr}(q) = 8q$.

- What is the short-run equilibrium price and quantity in this market?
- In the short-run equilibrium, how much does each firm produce? How many firms are in this industry?
- What is the total short-run producers surplus (i.e., profit) in this competitive market?
- Calculate the demand elasticity at the equilibrium price and quantity point.
- At this equilibrium point, if supply shifted inwards by a small amount, would market revenues go up or down?

Problem 75 Suppose that there are 100 firms operating smelters that each produce up to 1000 tons of aluminum. Ignoring the cost of the smelter (and just looking at short-run variable costs), the cost of producing aluminum (per ton) is \$250. Specifically, short-run opportunity cost is given by $C_{sr}^O(q) = 250q$ for any $q \leq 1000$. Thus, each firm is willing to supply 1000 tons of aluminum for any price greater than $p = \$250$, and they are unwilling to supply any output below a price of \$250.

- What do the (short-run) supply curves look like?
- If demand is given by

$$p = 1300 - 0.01 \cdot Q^d$$

what is the equilibrium price and quantity? What is each smelter's short-run economic profit (ignoring the cost of the smelter) at this price?

(c). Now suppose that demand unexpectedly shifts outward to

$$p = 1500 - 0.01 \cdot Q^d.$$

In the short run (given the number of smelters stays at 100), what is the new equilibrium price and economic profit of each firm?

(d). Suppose that the annual opportunity cost of a smelter is constant at \$50,000 per period. Thus, the long-run opportunity cost of smelting is $C_{lr}^O(q) = 250q + 50,000$ for $q > 0$ and $q \leq 1000$. (As a simplifying assumption, imagine that one can lease a smelter for this amount.) What market price of aluminum would be just sufficient to encourage a firm to buy/rent a smelter? Given this answer, what does the long-run industry supply curve for aluminum look like?

(e). Show/describe what happens to the industry short-run supply curves, the long-run supply curve and the market price when the market demand curve for aluminum unexpectedly shifts out from $p = 1300 - 0.01 \cdot Q^d$ to $p = 1500 - 0.01 \cdot Q^d$.

Problem 76 Suppose that in a market with perfect competition, the market demand function is given by $Q^d = 2200 - 100p$. Every firm in the industry has a short-run *opportunity* cost of production given by

$$C_{sr}^o(q) = 4q^2 + 2q + 4.$$

Note that the fixed cost of 4 is avoidable and not sunk.

(a). Compute each firm's short-run supply curve. [Be sure to indicate the range of prices for which the firms do not produce and to provide the formula for the quantity supplied in the price region in which they do produce.]

(b). If there are 800 firms, what is the short-run equilibrium price and quantity in this market?

(c). If each firm's long-run *opportunity* cost function is given by

$$C_{lr}^O(q) = 2q^2 + 8,$$

and there are an unlimited number of firms available with this cost function, what is the long-run equilibrium price in the market? How many firms will be producing?

Problem 77 Consider a constant-cost industry which is perfectly competitive market and comprised of homogeneous firms. Each firm has costs given by $C(q) = 10q$ (up to a capacity of 1 unit) and production is limited to 1 unit per firm. (Think of each firm representing one plant of 1-unit capacity; mergers are not allowed and technology requires 1-unit capacity, so firms never have any choice over scale.) In the long run, any firm can enter the industry for a per-period capital cost of \$75.

Timing: at the beginning of each period, each firm must make a planning decision (long-run decision) regarding whether or not to be in the market in the next period (i.e., whether or not to buy or sell capital). Each firm that has capital in place may also decide how much to produce in the current (e.g., short-run) period.

There are two possible demand curves in any period: low demand (given by $P(Q) = 110 - 2Q$) and high demand (given by $P(Q) = 310 - Q$). Each occurs with equal probability and these events are uncorrelated over time (i.e., high demand today tells us nothing about whether or not demand will be high tomorrow; high demand still arises with probability $\frac{1}{2}$). The realization of demand, however, will only be learned after capital decisions are sunk for that period.

(a). Suppose that initially there are $N = 100$ firms in the market. Calculate the short-run equilibrium price if demand is low, p_l , and the short-run equilibrium price when demand is high, p_h .

[Hint: it may be helpful to draw the industry's short-run supply function and the two different market demand curves in the same graph.]

(b). Given the current number of firms is $N = 100$, is each firm's expected short-run profit greater or less than the per-period cost of capital?

(c). How many firms will be in the industry if long-run expected economic profit is equal to zero? (I.e., what is the long-run equilibrium number of firms?) [Hint: Do not attempt to draw a long-run industry supply curve. Focus instead on the key condition of long-run equilibrium.]

Problem 78 In the competitive Yak butter industry in outer Bork, there are 10 identical, constant cost firms. Each has a long- and short-run marginal cost, MC, given by the equation:

$$MC_{sr} = MC_{lr} = 40 - 12q + q^2,$$

where q is the quantity produced by *the firm*. For each firm, the long run average cost is minimized at $q = 12$ units of Yak butter and short-run average cost is minimized at $q = 9$ units of Yak butter.

Market demand for the product in both the short and long run in this industry is given by

$$p = 160 - Q,$$

where Q is the total *market* output demanded and p is the price per unit.

(a). Explain why we know that the short-run average opportunity cost is $AC_{sr}^O = 13$ (i.e., $AVC=13$) when $q = 9$ and that the long-run average cost is $AC_{lr} = 40$ when $q = 12$.

(b). Using the the information above, is the industry in long-run equilibrium when the price is $p = 40$? Explain.

The following new fact is relevant for parts (c)-(d): As a result of a new free trade policy in Outer Bork, foreign sellers are now selling unlimited amounts of Yak butter at a price of 20 per unit.

(c). Are the domestic Yak butter firms going to shut down in the short-run? Explain.

(d). Analyze the long-run effects on the domestic Yak butter industry. Explain.

Problem 79 The industry demand for widgets is given by $Q = 240 - 10P$. Initially there are ten plants producing widgets. Each plant belongs to a different firm. (Indeed, there is a law restricting each firm to one plant.) Each has a cost function $16 + q^2$ where q is the number of widgets produced by each plant.

(a). Assuming initially that only these ten firm/plants may produce widgets, what will be the equilibrium price and quantity of widgets produced by each firm?

(b). If other firms may open a (single) plant and produce widgets, what will be the equilibrium price of widgets?

(c). How many widgets will each plant produce?

(d). How many firms will be producing widgets?

Problem 80 True, False, or Uncertain? [Explain your answer!] *In the short-run in a perfectly competitive industry with homogeneous firms, each firm will attempt to produce at the point at which its short-run average opportunity cost (i.e., $AC_{sr}^o(q) = AVC(q)$) is minimized.*

Problem 81 Consider a competitive industry which currently has 20 firms. There are 10 efficient firms and 10 inefficient firms. All firms have capacities of 1000 units; no output may be produced above capacity.

The efficient firms have $MC_{sr} = MC_{lr} = 100$, $AC_{sr}^O = AVC = 100$ and $AC_{lr}^O(q) = 100 + \frac{50,000}{q}$. Because the production efficiency of the low-cost firms is from learning and plant-specific process innovations, the technology is not transferable.

The inefficient firms have $MC_{sr} = MC_{lr} = 200$, $AC_{sr}^O = AVC = 200$ and $AC_{lr}^O(q) = 200 + \frac{50,000}{q}$. Hence, you can think of the inefficient firms having the same fixed costs of capital but different productivities from that capital.

Outside the industry there is a sufficiently large number of firms with the inefficient technology who represent the pool of potential entrants. Input costs are constant, regardless of industry size.

(a). Plot the short-run and long-run industry supply curves.

(b). If the industry is currently in long-run equilibrium and both types of firms supply output in equilibrium, what is the current market price?

(c). Suppose that the industry was in long-run equilibrium, but that demand suddenly shifted outward because of a *temporary* change in the economy. (Everyone knows the curve will shift back next period, however.)

1. What will happen to the short-run price (will it increase or decrease above the price in (b)) and the new long-run price (will it increase or decrease above the price in (b))?
2. Next period, who will enter or exit?
3. In the short-run, who will produce and how much will be produced?
4. Who will make short-run economic profits?

Problem 82 Suppose that long-run and short-run demand are identical (that is, both are represented by the same demand curve), but that short-run supply is more inelastic than long-run supply. Suppose also that all firm have identical cost structure and that the minimum short-run average opportunity cost function, $\min AC_{sr}^O(q)$ is greater than zero (i.e., the minimum AVC for each firm is greater than zero).

(a). Draw the three curves in one graph – demand and both short-run and long-run supply – when the market is in long-run equilibrium.

(b). Suppose that the demand curve shifts inward both unexpectedly and permanently. Describe the resulting changes of price over time – both the immediate change and the long-run change.

Problem 83 The industry demand for widgets is $Q = 600 - 10p$, where Q is the sum of all firms' production. Initially there are 40 plants producing widgets. Each plant belongs to a different firm. (A law restricts each firm to one plant.) Each plant has short-run economic costs equal to $C_{sr}^o(q) = 27 + 3q^2$ where q is the number of widgets produced by each plant. Short-run marginal costs for each plant are given by $MC_{sr}(q) = 6q$ and short-run average opportunity cost is $AC_{sr}^o(q) = 27/q + 3q$. These two curves intersect at $q = 3$.

(a). Assuming initially that only these 40 firms/plants may produce widgets and that the cost functions given above represent short-run economic cost, determine the short-run *industry* supply curve.

(b). (1) What is the equilibrium price in this market? (2) How many widgets will each firm produce? (3) What will be each firm's profits?

(c). Now assume that other firms may open a (single) plant and produce widgets if they wish in the long run. Also assume that the short-run and long-run cost functions are identical and all firms have access to this technology. Thus, long-run marginal and average opportunity costs are given by $MC_{lr}(q) = 6q$ and $AC_{lr}^o(q) = 27/q + 3q$. [Hence, the only distinction between the short-run and the long-run is that entry takes a period of time to occur.]

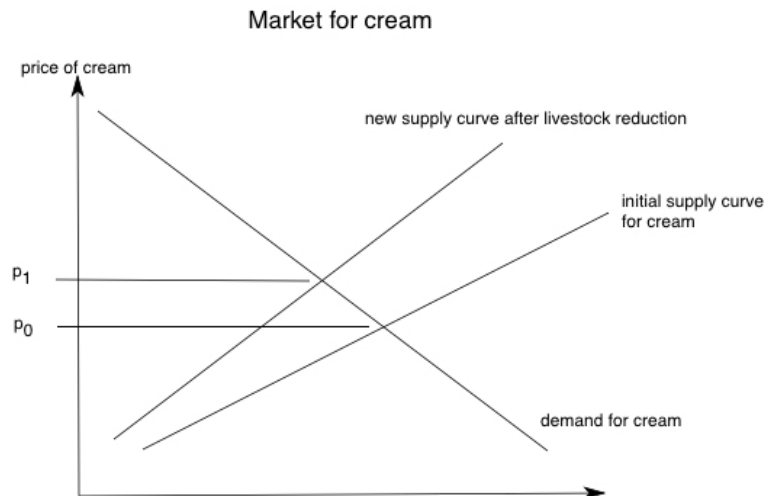
(d). (i) What is the equilibrium price in this market? (ii) How many widgets does each firm produce in the long run? (iii) How many firms are producing widgets? (iv) What is the profit of each firm?

Problem 84 True, False, or Uncertain. [Explain your answer!] *If all firms in an industry have the same cost conditions (i.e., identical technology, etc.), an increase in demand for the industry's product will not increase the price in the long run.*

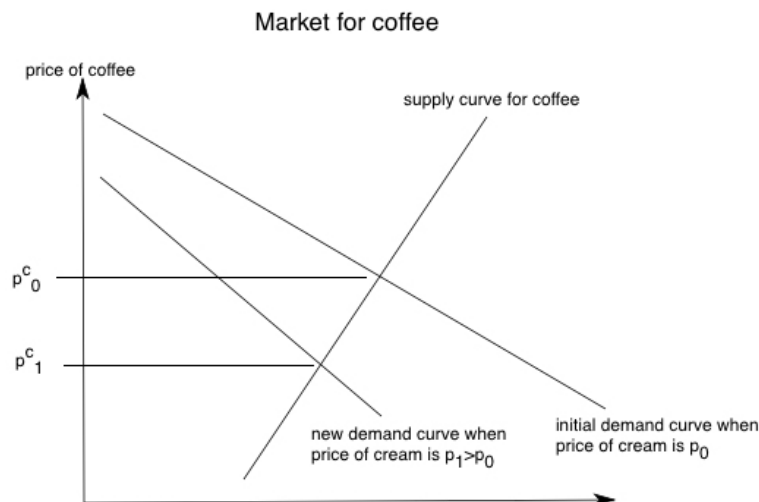
Problem 85 True, False, or Uncertain? [Explain your answer!] *In the short-run in a perfectly competitive industry with homogeneous firms, each firm will attempt to produce at the point at which its short-run average opportunity cost (i.e., $AC_{sr}^o(q) = AVC(q)$) is minimized.*

Answers to Week 1 Problems

1 True. We need to consider each of the two markets. First, a reduction in the supply of cream is equivalent to saying that the supply curve for cream shifts inward. Such a shift implies that the price of cream will increase. In the figure, the price of cream moves from p_0 to $p_1 > p_0$.



Second, consider the impact of an increase in the price of cream in the coffee market. Because coffee and cream are demand complements, the increase in the price of cream implies that the demand curve for coffee will shift inward. An inward shift in coffee demand implies that price decreases in the coffee market. In the figure, the price of coffee moves from p_0^c down to p_1^c .



2 (a). Oil and gas are substitutes because the price of oil is positively related to the demand for natural gas in the demand equation: $\frac{\partial Q_g^d}{\partial p_o} > 0$.

(b). With a price of $p_o = 8$, the demand and supply relationships become

$$Q_g^s = 16 + 2p_g,$$

$$Q_g^d = 30 - 5p_g.$$

Solving for equilibrium price, $16 + 2p_g = 30 - 5p_g$, and so $p_g = 2$.

(c). The equilibrium price and quantity in (b) is $p_g^* = 2$. The equilibrium quantity of gas that is traded can be determined from either the supply or demand function. Using supply, we have $Q^* = 16 + 2(2) = 20$ (or alternatively, using demand, $Q^* = 30 - 5(2) = 20$). The slope of the supply function, $\Delta Q_g^s / \Delta p_g = 2$, so the elasticity of supply is

$$\varepsilon_g^s = (2) \frac{2}{20} = 0.20.$$

(d). With a price of $p_o = 16$, the demand and supply relationships become

$$Q_g^s = 18 + 2p_g,$$

$$Q_g^d = 60 - 5p_g.$$

Solving for equilibrium price, $18 + 2p_g = 60 - 5p_g$, and so $p_g = 6$. The price triples in response to a doubling of the price of oil.

3 (a). Set $Q^d = Q^s$ to determine price.

$$1500 - 100p = 500 + 300p,$$

$$1000 = 400p,$$

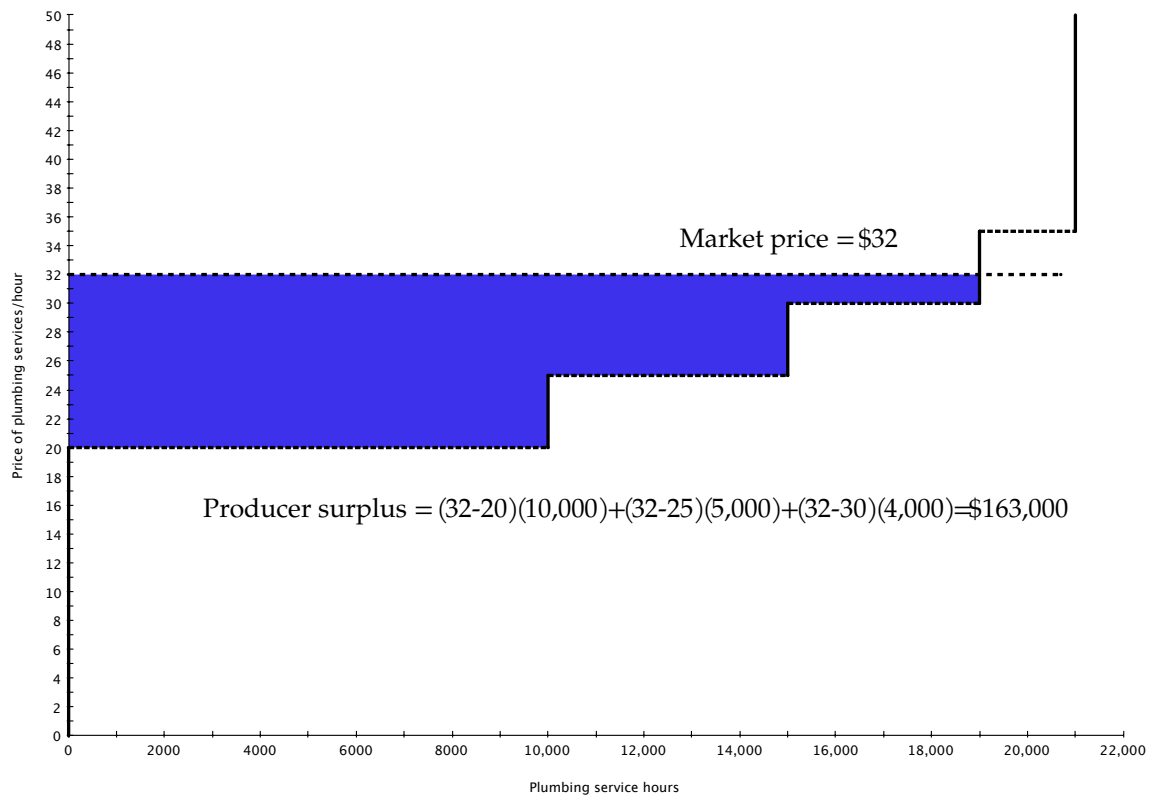
$$p^* = \$2.5.$$

Obtain Q by substituting into either expression: $Q^* = Q^d = 500 + 300(2.5) = 1250$.

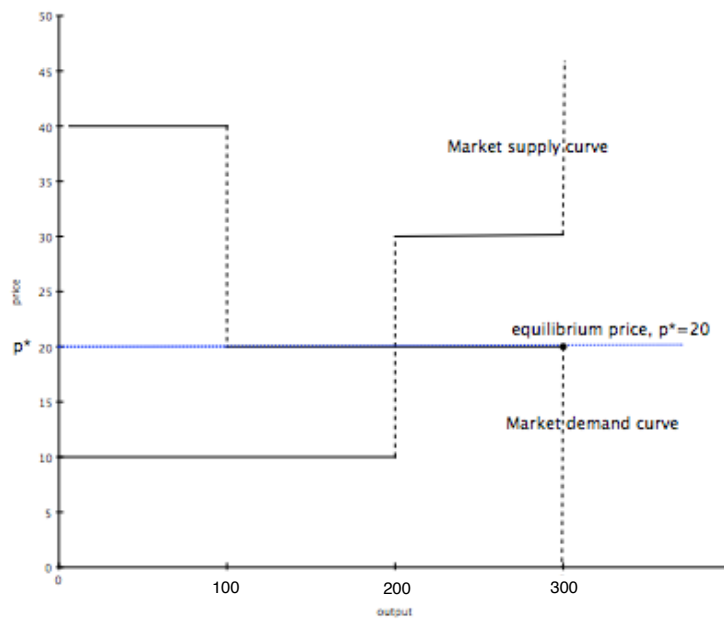
(b). Price elasticities. For demand, $\epsilon^d = (-100) \frac{2.5}{1250} = -\frac{250}{1250} = -0.2$. For supply, $\epsilon^s = (300) \frac{2.5}{1250} = \frac{750}{1250} = 0.6$

(c). At $p = 3$, $Q^d = 1500 - 100(3) = 1200$ and $Q^s = 500 + 300(3) = 1400$. Thus, there is a surplus supplied by the market equal to $Q^s - Q^d = 200$ thousand bushels that the government would be required to support a \$3 price per bushel.

4 See graph below for calculation.

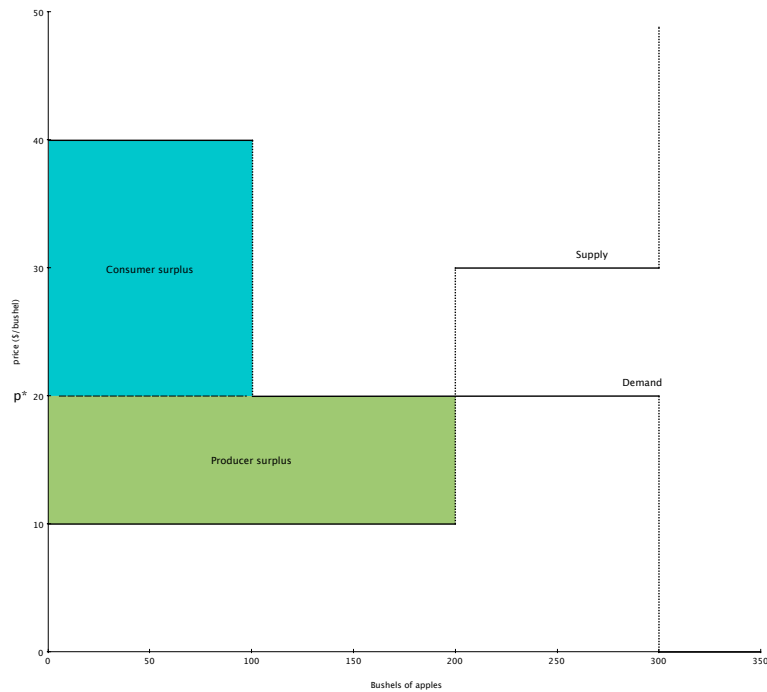


5 (a). The demand and supply curves are depicted in the figure below.



At an equilibrium price of $p^* = 20$, 200 bushels of apples are supplied by the sellers and the buyers are willing to buy $Q^* = 200$ bushels. Hence, it is an equilibrium. [Note, because the demand curve is perfectly flat, we have the strange situation where the buyers are indifferent to buying any amount between 100 and 300 bushels. With a strictly decreasing demand curve, this indifference would disappear.]

(b)-(c). The consumer and producer surplus are illustrated in the following graph.



$$CS = (100 \text{ bushels})(40\$/\text{bushel} - 20\$/\text{bushel}) = \$2000,$$

$$PS = (200 \text{ bushels})(20\$/\text{bushel} - 10\$/\text{bushel}) = \$2000.$$

(d). If the government taxed seller's \$5 per bushel, the effect would be that the supply curve would shift upwards by exactly \$5. Hence, the equilibrium pre-tax price would remain at \$20; consumers would pay \$20 for a bushel of apples and consumer surplus would remain at $CS = \$2000$. Producers would receive \$20 from the consumers but would have to pay \$5 to the government, so the after-tax receipt is only 15\$/bushel. As a consequence, producer surplus decreases to

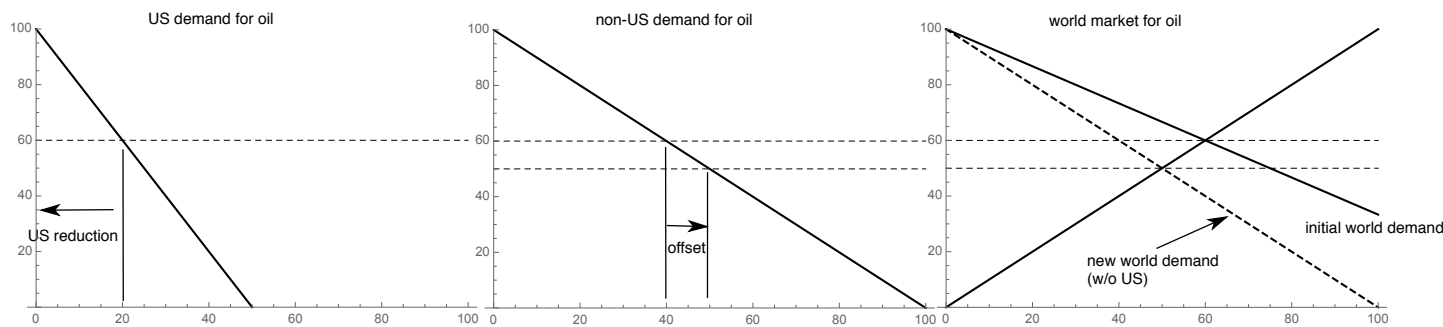
$$PS = (200 \text{ bushels})(20\$/\text{bushel} - 15\$/\text{bushel}) = \$1000.$$

As we will prove in week 2 of the course, it does not matter if the government decided to shift the tax to the buyers. In equilibrium, the producers would still pay for the tax out of their producer surplus and the consumers would not be adversely impacted.

6 The key to answering this question is to understand that the US policy will reduce the US consumption of oil, but this reduction of oil demand will also reduce the world price of oil. Such a price reduction will indirectly increase the oil consumption of non-US consumers, which will offset the direct US reductions. The result may be that world consumption of oil will fall somewhere between 0 and 20 million barrels of oil per day.

A critical factor is the world supply curve for oil. If it is vertical (all wells pumping to capacity but no new drilling), the same amount of oil will be consumed after the US reduction because the non-US consumers will increase their consumption to exactly offset US reductions. On the other hand, if the supply curve is horizontal (not realistic), then the price will not decrease when US consumers no longer buy oil. Without a decrease in price, non-US consumption of oil will remain unchanged and the billion tons in carbon reduction will result. Note that the nature of the rest of the world's demand curve is also relevant to this answer. If the demand from the rest of the world is perfectly elastic (horizontal demand), then the non-US consumers will buy up all of the oil previously purchased by the US and the US policy will have no effect on the world consumption of oil. If the demand curve for the rest of the world is vertical, then the price falls but no additional demand arises to offset US consumption and therefore world oil consumption is reduced by the full 20 million barrels per day.

I've drawn some simple graphs trying to illustrate this phenomenon. The panel on the left has the original US demand for oil. The center panel is the non-US demand for oil. (For convenience, I've assumed that both curves have the same intercept.) The third panel contains the sum of the two demand curves (world demand) and world supply. I have also drawn the non-US demand in the third; once the US demand for oil is eliminated, this becomes the world demand. The World price of oil falls, but the quantity consumed does not fall by the full 20 million barrels per day because of the offsetting non-US increase in consumption.



Aside: You could also solve this problem with algebra and calculus, but it is quite challenging (i.e., something I'd only expect from PhD students in economics). That said, I know a few of you like challenges and are comfortable with calculus, so here is the purely mathematical answer.

Let $D_r(p)$ be the rest of the world's demand for oil at p ; let $S(p)$ be the world supply of oil;

and let x equal the US demand for oil at the original equilibrium price of p_1 ; and let p_2 be the new equilibrium price once the US demand is withdrawn from the market.

Define $p(x)$ from the supply=demand relationship

$$D_r(p(x)) + x = S(p(x)),$$

or

$$S(p(x)) - D_r(p(x)) = x.$$

By definition, $p(x) = p_1$ and $p(0) = p_2$. We want to determine dQ/dx for small x .

Differentiating the above equation, we have

$$p'(x)(S'(p(x)) - D'_r(p(x))) = 1,$$

and thus

$$p'(x) \frac{S(p(x))}{p(x)} \left(S'(p(x)) \frac{p(x)}{S(p(x))} - D'_r(p(x)) \frac{p(x)}{S(p(x))} \right) = 1,$$

or more succinctly

$$p'(x) \frac{S(p(x))}{p(x)} (\varepsilon^s - \varepsilon^d) = 1.$$

Thus,

$$\frac{dQ}{dx} = p'(x)S'(p(x)) = \frac{p(x)}{S(p(x))} \left(\frac{1}{\varepsilon^s - \varepsilon^d} \right) S'(p(x)) = \frac{\varepsilon^s}{\varepsilon^s - \varepsilon^d}.$$

We conclude that there is no change in the world consumption of oil (full offset) if either supply is perfectly inelastic or demand is perfectly elastic; and there is a one-to-one reduction (no offset) if supply is perfectly elastic or demand is perfectly inelastic.

7 The billionaire will reduce the world's consumption of mahogany, but not by as much as 100 million cubic feet. There are two equally valid ways to view this.

(1). Supply shift. You can plot the original supply and demand curves with the horizontal axis representing "acres of rainforest cleared for mahogany". With this graph, we would model the environmentalist as shifting the available supply of rainforest land for lumber to the left horizontally by exactly 100,000 acres. The new equilibrium price is higher than before and the quantity of rainforest sold for lumber decreases, but not by as much as 100,000 acres because the higher price encourages more land to be supplied for lumber which offsets the 100,000.

(2). Demand shift. Alternatively, you can plot the original supply and demand curves with the horizontal axis representing "acres of rainforest sold", which includes rainforest sold for lumber AND rainforest sold to the environmentalist. With this graph, we would model the environmentalist as shifting the demand for rainforest land to the right horizontally by exactly 100,000 acres. The new equilibrium price is higher than before (and is the same as above) and the quantity of rainforest sold increases. Suppose it increases by 40,000. Then this implies that 100,000-40,000=60,000 fewer acres were cleared for lumber because of the higher price.

In either case, some of the 100 million cubic feet of mahogany that is removed from the market by the billionaire will be replaced by some sellers at the new higher price. The net effect is less than 100 is saved.

8 (a). Soybeans are a substitute because the coefficient of p_{sb} is positive in the demand for wheat equation. hence, as the price of soybeans increases, people substitute into wheat.

(b). Because corn is also a substitute for wheat, the demand for wheat increases at any given price, so the demand curve must shift outward.

(c). First, substitute into the demand relationship the given prices. We have

$$Q_w^s = 40 + p_w,$$

$$Q_w^d = 4(50) + 7(40) - 10p_w = 480 - 10p_w.$$

Solving for p_w , we have

$$40 + p_w = 480 - 10p_w,$$

so $11p_w = 440$ and $p_w^* = 40$. Hence, $Q_w^* = 80$.

(d). The point elasticity of demand is given by

$$\epsilon_w = \frac{dQ_w}{dp_w} p_w Q_w = (-10) \frac{40}{80} = -5.$$

9 Equilibrium occurs at $30 - p = 5 + 4p$. Hence, $p^* = 5$ and $Q^* = 25$. The point price elasticity of demand is thus $(-1)(5/25)=-0.2$. The point supply elasticity is $(4)(5/25)=0.80$.

10 False. Gateway computers have many very close substitutes – other Intel-based personal computers, so we should expect the elasticity of demand to be much higher.

11 (a). Substituting in for $p_{wine} = 20$ and $I = 10$, we have

$$Q_{beer}^d = 200 - 4p_{beer}.$$

Using $Q_{beer}^s = 100 + 6p_{beer}$, we have

$$200 - 4p_{beer}^* = 100 + 6p_{beer}^*.$$

Hence, after a bit of algebra, $p_{beer}^* = 10$. The corresponding quantity is $Q^* = 100 + 6(10) = 160$.

(b). (i). The income elasticity is

$$\epsilon_I = \frac{\partial Q^d}{\partial I} \frac{I}{Q} = (2) \frac{10}{160} = \frac{1}{8} = .125.$$

(ii) It is a normal good; (iii) but not a luxury good.

12 (a). First, note that the supply curve will shift to the left, so the movement in price and quantity is along the demand curve, hence the elasticity of demand is the appropriate price elasticity to consider. Second, we are told the percentage change in quantity (not price!), so to find the percentage change in price, we must solve the following simple equation:

$$\varepsilon_d \times (\Delta\%p) = \Delta\%Q,$$

or more simply $(-0.5)(\Delta\%p) = 10$, so $\Delta\%p = 20$ percent.

(b). Price increases, quantity decreases and demand is inelastic. Hence, revenues increase using our relationship derived in class.

(c). If demand decreases, then both price and quantity decreases, and so revenue must decrease.

13 False. If demand decreases at every price (i.e., the demand curve shifts inward), then equilibrium price decreases but revenue also decreases. [Note that this is a movement along the supply curve, and not the demand curve, so the demand elasticity is irrelevant.]

14 True. An inward supply shift (caused by the bad weather) will raise price and reduce output. Since consumer surplus is the area under a demand curve and above the price, this area necessarily decreases. The only possible relevance of the elasticity is to rule out the case in which the demand curve is horizontal, and hence there was no consumer surplus to begin with. Other than the fact that $\varepsilon_d \neq -\infty$, the elasticity of demand is not needed to answer the question.

The demand elasticity is irrelevant for this question. It is true that because the demand is elastic, a supply shift inward will raise the price and therefore reduce consumer expenditures (and revenue), but this is not a direct measure of surplus. Indeed, expenditures by consumers go down but so does consumer surplus.

15 Revenues must increase. When the supply curve shifts right, the market price must decrease. If the demand curve is elastic, then a decrease in the price must increase revenues (i.e., total market expenditures).

16 False. There are at least two reasons for this. First, assuming that these changes in prices and revenues occur due to shifts in the market supply curve, cigarettes are actually inelastic. This follows because revenues increase as the price increases. You could have backed out an average price from the revenues and unit sales, and then used the percent changes in these average prices to calculate the demand elasticity, but this was entirely unnecessary to answer the question. Second, it is unclear that whether the demand curve is not also shifting. Hence, the two data points data are from two different curves, so we cannot calculate the elasticity of demand without knowing more about the underlying cause and magnitude of the shifts.

17 (a). Setting demand equal to supply, we solve for the equilibrium price:

$$10,000 - 200p = 1200p - 18,000,$$

which simplifies to

$$28,000 = 1400p,$$

and thus $p^* = \frac{28,000}{1400} = 20$. At this price $Q^* = 10,000 - 200(20) = 6000$.

(b). The elasticity of demand is

$$\varepsilon^d = \frac{dQ^d}{dp} \frac{p}{Q} = (-200) \frac{20}{6000} = -0.67.$$

Hence demand is inelastic.

(c). Note that the choke price for demand is $p = 50$ and the minimum price for supply is $p = 15$. Thus,

$$CS = \frac{1}{2}(50 - 20)(6000) = 90,000,$$

$$PS = \frac{1}{2}(20 - 15)(6000) = 15,000.$$

(d). Setting demand equal to supply, we solve for the equilibrium price:

$$10,000 - 200p = 800p - 12,000,$$

which simplifies to

$$22,000 = 1000p,$$

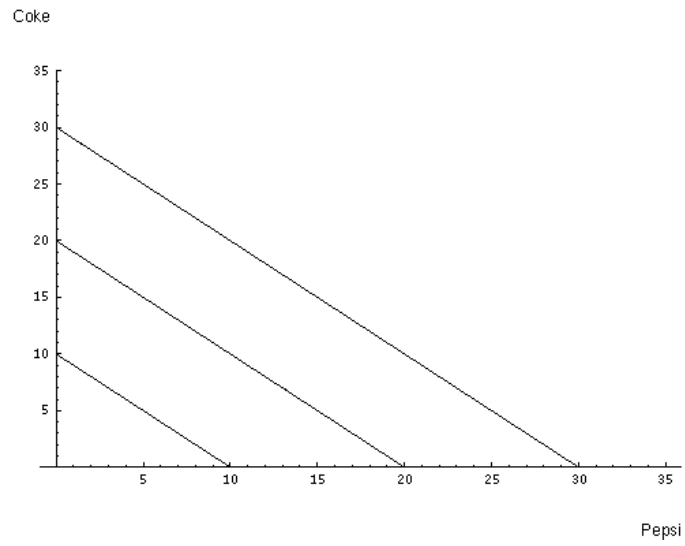
and thus $p^* = \frac{22,000}{1000} = 22$. At this price $Q^* = 10,000 - 200(22) = 5600$. At this price and quantity, the demand elasticity is

$$\varepsilon^d = \frac{dQ^d}{dp} \frac{p}{Q} = (-200) \frac{22}{5600} = -0.79.$$

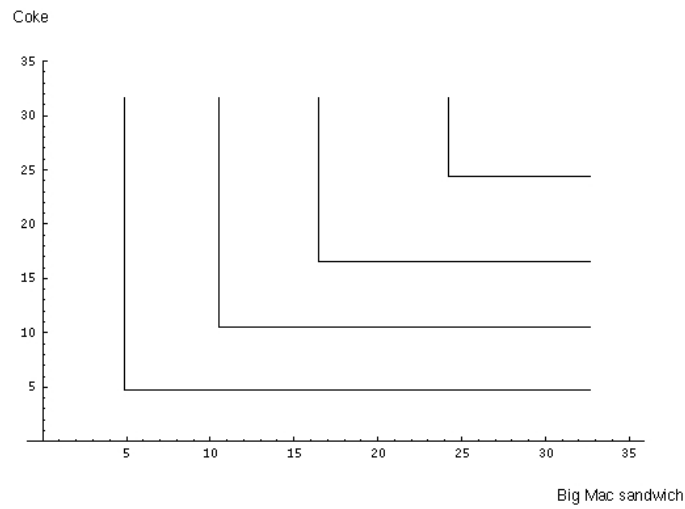
Thus, demand is still inelastic.

(e). Computing this directly, we have the original revenue of $R = 20(6000) = 120,000$ and the new revenue of $R = 22(5600) = 123,200 > 120,000$, and thus revenues increased. Given the equilibrium price increased from $p = 20$ to $p = 22$ along the inelastic region of the demand curve, we know that revenues must increase without doing the direct computations.

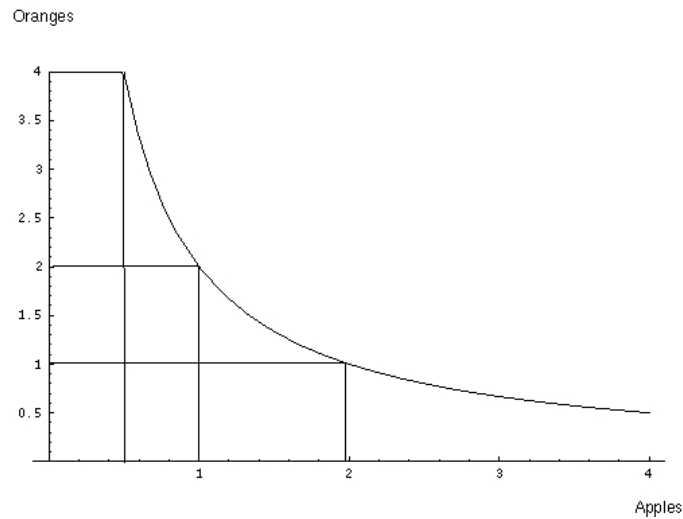
18 (a). In the Coke-Pepsi case, the indifference curves are straight, downward sloping lines with a 45-degree slope.



(b). In the Coke-Big Mac case, the indifference curves are L-shaped, with the angle occurring at a 45-degree line through the origin.



(c). These indifference curves are rectangular hyperbolas of the form $(\text{Apples}) \times (\text{Oranges}) = \text{Constant}$



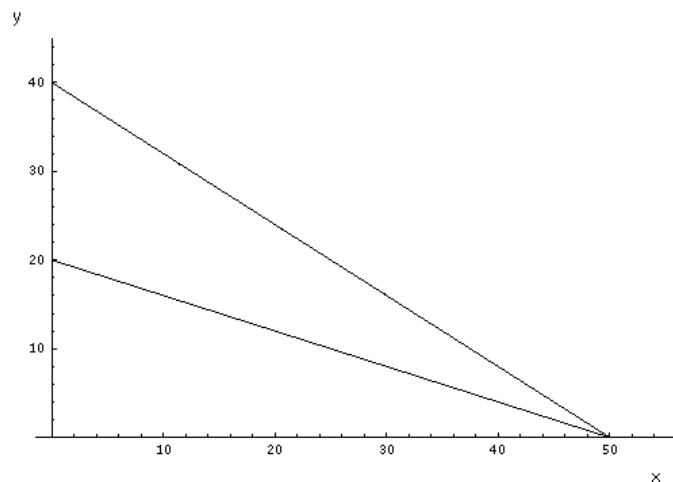
19 FALSE. This utility function is linear, and thus the MRS remains constant throughout. [Remember gasoline varieties?] If you actually differentiated this function to obtain the MRS, you found

$$MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{2}{3},$$

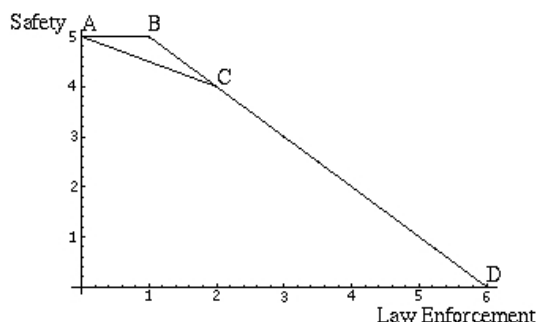
which is independent of x and y .

20 (a). The budget constraint has a y intercept of $200/10 = 20$, and x intercept of $200/4 = 50$, and a slope equal to $-p_x/p_y = -2/5$.

(b). The new budget constraint has a y intercept of $200/5 = 40$, the same x intercept of $200/4 = 50$, and a new slope of $p_x/p'_y = -4/5$.



21 (a) Program 1: refer to the line segment BCD in the diagram below;
 Program 2: refer to the line segment ACD below. Each axis is measured in millions of Euros.



(b) As can be seen from the diagram, the programs are identical in the region CD. Clearly if you intended to be in this range, you are indifferent. In the range before that, the budget set for program 1 is larger than that for program 2; hence you should prefer program 1.

22 FALSE. It is not possible to tell without knowing the prices of pizza and beer. In general, one could choose prices and income in order to make any allocation of pizza and beer optimal.

23 TRUE. If all prices (and income) but beer increase by 10 percent, it is *as if* the price of beer decreases by 10 percent. Assuming that the law of demand holds, if the price of beer decreases, consumption will increase. It would also have been correct to have answered UNCERTAIN if you entertained the possibility of an upward-sloping demand curve (e.g., a Giffen good).

24 True. The marginal utility per dollar spent on strawberries is $\frac{MU_s}{p_s} = \frac{4}{2} = 2$; the marginal utility per dollar spent on chocolate is $\frac{MU_c}{p_c} = \frac{7}{3} > 2$. Therefore, Robert should consume chocolate for dessert. Alternatively, $MRS_{s,c} = \frac{MU_s}{MU_c} = \frac{4}{7} < \frac{p_s}{p_c} = \frac{2}{3}$, and so again chocolate is preferred.

25 FALSE. At the optimum, $\frac{MU_p}{p_p} = \frac{MU_s}{p_s}$. At $p_p = 2.50$, this implies

$$\frac{MU_p}{p_p} = \frac{9}{2.50} \neq \frac{MU_s}{p_s} = \frac{15}{5} = 3.$$

Since $\frac{9}{2.5} \neq 3$, p_p should not be 2.50. In this case for the optimum to hold, $p_p = 3.00$.

26 (a). They are perfect substitutes because the marginal rate of substitution is constant.

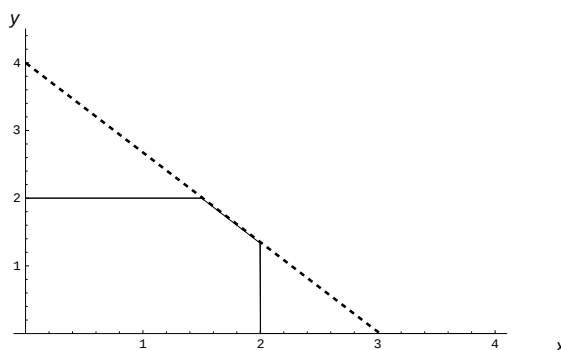
(b). At an interior equilibrium, $MRS_{wine,beer} = \frac{p_w}{p_b}$. But in this case, it is always true that $MRS_{w,b} = 3 < \frac{p_w}{p_b} = 3.5$. Thus, should consume only beer (note that a straight line indifference-curve implies a corner solution - refer to diagram). The optimal ratio of wine to beer is 0.

27 FALSE. At the optimum, MRS equals the price ratio. In our case, since both prices are increasing in the same proportion, the price ratio remains unchanged. Hence, the value of MRS at the optimum also remains unchanged, even though possibly different quantities and proportions of x and y will be consumed.

28 FALSE. Demand for good A must be price elastic. If the consumer's expenditures on good B rises, it must be that the consumer's expenditures on good A have fallen (total expenditures on the two goods must add up to the total income of the budget constraint.) If expenditures on good a FALL when the price of A rises, demand for good A must be price elastic.

29 TRUE. The marginal rate of substitution for Dominique at her current consumption level is $MRS_{aus,french} = \frac{120}{300} = \frac{2}{5}$, while the price ratio is $\frac{10}{25} = \frac{2}{5}$, and so providing her preferences exhibit decreasing MRS (which seems reasonable), she is maximizing utility. You could have said UNCERTAIN due to the fact that I did not state that her preferences were convex (e.g., there could have been increasing marginal rates of substitution in some region), but you should have noted the equality of the price ratio and MRS is consistent with optimal choice. Both answers received full marks.

30 (a).



(b). Since the downward sloping portion of your budget constraint has slope of $-\frac{4}{3} = -1.33$ and your $MRS_{x,y} = \frac{240}{200} = 1.2$, you will want to consume all y (since the market's price of x in terms of y is greater than your personal relative valuation). Unfortunately, you will run into the government constraint at $y = 2$, so you will choose the kink or corner at $y^* = 2$ and then consume the remainder of your income on x : i.e., $x^* = \frac{(I - p_y y^*)}{p_x} = \frac{60 - 15(2)}{20} = 1.5$.

It was not enough to get the correct answer by luck or guessing; you needed to get with the appropriate economic arguments for full credit.

31 True. (This was probably the hardest question on the exam.) The reasoning is subtle and requires a few logical steps. It is easiest to see by considering that the statement is false.

Suppose to the contrary that both goods are demand substitutes but that at least one of the goods is inelastic. For the sake of argument, let's say that x is an inelastic good. Step 1: if x is inelastic, as the price of x increases, the total expenditure spent on x must also increase. Step 2: this then implies that the total expenditure on y must decrease (as income is fixed), and hence the demand of y must also decrease (since p_y has not been changed). Conclusion: combining the two steps of logic, we have that a price increase in x is followed by a decrease in demand of y . This conclusion contradicts the statement that the goods are demand substitutes. Hence, the assumption that the statement is false is incorrect. Both goods must be elastic.

32 False. If consumption increases with income, the good is normal. It is not relevant that the good increases in demand by a lower proportion than income; all that matters is that it increases.

33 The \$15 season ticket is better. First, consider the plan which requires a \$1 charge for admission. Under this plan, each resident will go to the pool $q = 18 - 3(1) = 15$ times per year. Note that the demand curve intersects the price axis at $p = 6$, so it follows that the per capita consumer surplus is given by $CS = (1/2)(15)(6 - 1) = 37.50$.

In contrast, under the season pass plan, each of the residents will go $q = 18 - 3(0) = 18$ times a year and obtain a per-capita consumer surplus of $CS = (1/2)(18)(6) = 54$, ignoring the cost of the season pass. With the additional season pass cost of \$15, each consumer will obtain a net surplus of $54 - 15 = 39$.

Both plans raise the same amount of contribution of \$15 per consumer, but the season pass plan leaves more consumer surplus.

34 She should take the offer. She is indifferent – on the margin – between a compact disk and 5 pounds at her optimal choice of 40 CDs. The new offer will take her to the same point on the indifference curve, but importantly, it will also allow her to substitute 4 pounds for each additional compact disk. She will therefore accept the offer and purchase more disks, moving to a higher indifference curve. Only if the additional CD causes the indifference curve to become flatter than -4 will she stay at her current consumption. [Note that her indifference curve at 40 CDs has slope -5 so this assumption seems unwarranted; with continuous consumption units and smooth indifference curves she will always buy at least slightly more than her current allocation.]

35 (a)-(b). If your reduction in demand is 5 percent and the change in price is 10 percent, then the elasticity of demand is

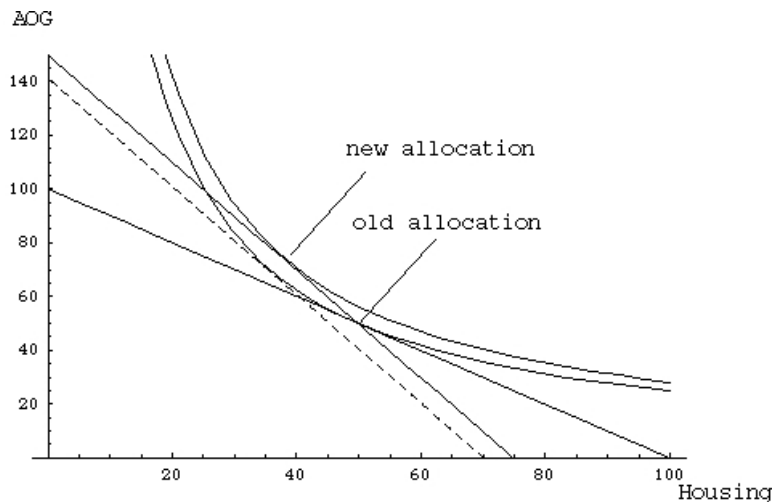
$$\varepsilon = \frac{-5\text{percent}}{10\text{percent}} = -\frac{1}{2}.$$

Hence, demand is inelastic. It follows that expenditures on apples must increase (part (b)).

Alternatively, you could have obtained this by noting that your expenditures on apples must increase following the price increase because price increased more in percentage terms than quantity decreased; hence, demand is inelastic.

(c). Given your expenditures on apples increases, you have less money left over to spend on all other goods. Remember, income hasn't changed. Hence, your expenditures on AOG must decrease. Because the prices of other goods have also not changed, you must be reducing your consumption of AOG.

36 It is true that your COLA allows them to continue to buy the same bundle of goods. But now that housing is more expensive, your transferred workers will prefer to adjust their bundle of goods to something with less housing. This adjustment raises their utility so they are economically happier than if they stayed in Chicago. A figure can easily demonstrate what's going on here.



The old and new budget lines go through the same bundle of goods (by construction), but the new budget line is steeper (because housing is more expensive). Here, I assumed an initial income of \$100,000 and a housing index of 1 for Chicago (i.e., the price ratio of housing in Chicago to other goods is set to 1). Then I consider a new budget line which contains the original point of consumption, but where the housing index is now twice as high. Before, your employee consumed \$50,000 per year in housing; given the increased housing cost in San

Francisco (the same housing would cost \$100,000 per year), you decide to raise the salary by \$50,000. As you can see, the consumer can move into other goods and out of housing and increase utility. In short, your COLA plan should have attempted to mimic the substitution effect by itself (i.e., stay on the same indifference curve given the new relative prices). In my graph, you would have needed to only increase the salary by \$40,000 to this end.

37 (a). $E[\tilde{x}] = \phi_1 x_1 + \phi_2 x_2 + \phi_3 x_3 = \frac{1}{2}(-10) + \frac{1}{4}(15) + \frac{1}{4}(25) = 5.$

(b). $E[\tilde{y}] = \phi_1 y_1 + \phi_2 y_2 + \phi_3 y_3 = \frac{1}{2}(-25) + \frac{1}{4}(25) + \frac{1}{4}(45) = 5.$

(c). If the risk averse person has mean-variance preferences, then it is straightforward to see both assets have the same expected value, but $\tilde{x}$ has a lower variance than $\tilde{y}$, hence a risk averse (i.e., variance averse) person would choose asset $\tilde{x}$. A more general answer (and one that is beyond the lecture material) is to note that a risk averse person with a concave utility function (not necessarily just variance averse) would always choose $\tilde{x}$ over $\tilde{y}$. To see that $\tilde{x}$ is preferred, note that $\tilde{y}$ is equivalent to taking one unit of $\tilde{x}$ and purchasing second unit of $\tilde{x}$ paid at the price of its expected value: $y_1 = 2x_1 - 5$, $y_2 = 2x_2 - 5$, $y_3 = 2x_3 - 5$. A risk averse person would never buy a second risky asset at its expected value if it is perfectly and positively correlated with the initial risky asset.

38 (a). $EV = \frac{1}{4}(10,000) + \frac{1}{2}(0) + \frac{1}{4}(-10,000) = 0.$

(b). Consider his expected utility of taking the gamble:

$$EV = \frac{1}{4}u(110,000) + \frac{1}{2}u(100,000) + \frac{1}{4}u(90,000) = \frac{1}{4}30 + \frac{1}{2}26 + \frac{1}{4}20 = 25.5.$$

In contrast, the expected utility of 100,000 for sure is $u(100,000)=26$. Therefore, rejecting the project is better for Faruk.

(c). The project generates an expected utility of 25.5. We are told that \$99,000 also generates the same utility. Thus, $CE = 99,000$. It follows that $\rho = E[\tilde{x}] - CE = 100,000 - 99,000 = 1000$.

39 (a). $E[x] = .5(400) + .5(-200) = 100.$

(b). Solve $E[x] = \phi_G(400) + (1 - \phi_G)(-200) = 0$ for ϕ_G . The answer is $\phi_G = \frac{1}{3}.$

(c). Solve $E[x] = \phi_G(400) + (1 - \phi_G)(-100) = 0$ for ϕ_G . The answer is $\phi_G = \frac{1}{5}.$

(d). At this probability, a person without limited liability who would bear the full cost of the \$200 loss would have an expected value of $E[x] = \frac{1}{5}(400) + \frac{4}{5}(-200) = -80 < 0.$

(e). Because the company does not face the full effect of the downside risk, its payoff (as a function of the project outcomes) is convex. This causes a firm to behave as if they are risk loving.

40 FALSE. Joan has diminishing marginal utility of income, hence, her utility function is concave. This is an alternative definition of risk aversion, hence she is not risk loving.

41 This question is requires some subtle thinking.

(a). If both types of cars are sold, the expected value to a buyer is simply $EV = .5(5,000,000) + .5(1,000,000) = \$3,000,000$.

(b). Although the price of 2,500,000 is less than the expected value of the car *providing that both types of cars are sold*, you should intuit that only the owners of lemons would be willing to sell at that market price. That is, the supply curve for good cars is zero for any price less than 4,000,000; the supply curve for bad cars is 0 only for prices lower than 800,000. Thus, the quality of the car is bad, and the actual value to the buyer is only 1,000,000.

(c). At any price between 800,000 and 1,000,000, buyers and sellers of bad cars should be willing to transact. There is no price at which goods cars will be both supplied and purchased as long as buyers cannot distinguish the cars.

(d). This is related to the general problem of adverse (or self-) selection. Just as insurance markets may break down because only bad insurance risks will demand insurance at a given price, the used car market might break down because only bad suppliers will supply cars at a given price.

42 (a) $EV = 0.5(\$5,000) - 0.5(\$2,500) = \$1,250$.

(b) She is risk averse since her utility function is concave, which is equivalent to her having diminishing marginal utility of income. To see the concavity, you could have plotted several numbers and determined that the marginal gain from an additional dollar is decreasing. You also could have plotted the function. Alternatively, you could just note that the second derivative of the utility function with respect to income is negative.

$$Uu'(x) = \frac{d}{dx} \frac{1}{2} x^{-\frac{1}{2}} = -\frac{1}{4} x^{-\frac{3}{2}} < 0.$$

(c) Remember to add her original wealth to the projects profits/losses to get the final wealth in each state. Thus,

$$\begin{aligned} E[u(x)] &= 0.5\sqrt{5000 + 5000} + 0.5\sqrt{5000 - 2500} \\ &= 0.5\sqrt{10,000} + 0.5\sqrt{2500} \\ &= 0.5(100) + 0.5(50) = 75\text{utils} > 70.5\text{utils} = \sqrt{5000}. \end{aligned}$$

(d) Given the expected utility of the project is 75 utils, the sure income of I that just makes Elaine indifferent is $\sqrt{I} = 75$, or $I = (75)^2 = \$5,625$. Since her expected income (initial wealth plus expected profits) is $\$5,000 + \$1,250 = \$6,250$, she would be willing to pay $6,250 - 5,625 = \$625$ to remove the risk.

(e) You received full marks by explaining that Elaine's expected utility must be higher because of the diversification of risk via the partnership.

To see this, note that there are four possible outcomes for the pooled projects, each having $\frac{1}{4}$ probability chance of occurring: (i) Elaine's and Kramer's businesses both do well (shared profits are $(5000 + 5000)/2 = \$5000$); (ii) Elaine's and Kramer's businesses both do badly (shared losses are $(2500 + 2500)/2 = \$2500$); (iii) Elaine's business does well while Kramer's fails (shared profits are $(5000 - 2500)/2 = \$1250$); (iv) Elaine's project does badly while Kramer's succeeds (shared profits are $(-2500 + 5000)/2 = \$1250$).

Thus, expected utility is given by

$$\begin{aligned} E[u(x)] &= 0.25\sqrt{5000 + 5000} + 0.25\sqrt{5000 - 2500} + 0.25\sqrt{5000 + 1250} + 0.25\sqrt{5000 + 1250} \\ &= 0.25\sqrt{10,000} + 0.25\sqrt{2500} + 0.25\sqrt{6250} + 0.25\sqrt{6250} \\ &= 77.03 \text{utils.} \end{aligned}$$

If you actually worked out the expected utility in the partnership (which the question did not ask for) it is 77.03 utils, higher than before (75 utils).

(f) Moral hazard. Kramer may not work as hard as he should at the business, possibly resulting in the probability that the business succeeds falls below 0.5.

(g) Adverse selection. Particularly bad entrepreneurs may show up to pool projects. That is, individuals with a probability of failure that is close to 1 may show up, while individuals with a probability of failure close to 0 may prefer to stay by themselves rather than pool with Kramer and Elaine. It's a lot like Groucho Marx's joke about belonging to clubs.

43 True. This gamble has zero expected value.

$$E[x] = \frac{1}{2}2000 + \frac{1}{4}1000 - \frac{1}{4}5000 = 0.$$

Hence, a risk averse person would strictly prefer to avoid it and get zero for sure.

44 (a). He is risk averse since he prefers the certain income rather than the risky income which has the same expected value.

(b) The risk premium is the difference between the expected dollar value of the risk and the certain equivalent. In this case, it's the difference between $.5(360,000) + .5(0) = 180,000$ and $360,000 - 200,000 = 160,000$, which is $\$20,000$.

(c). Gail would probably not offer such insurance since Ellsworth could respond by working less diligently. [This is an example of moral hazard.]

45 (a). The formula for consumer surplus with demand $p = 10 - 2q$ is

$$\frac{1}{2}(q)(10 - p) = \frac{1}{2}q(2q) = q^2.$$

In the high-supply state, $p = 6$ so $CS = 4$ and in the low-supply state, $p = 8$, so $CS = 1$:

(b). The expected value of the market place is $E[CS] = \frac{1}{2}4 + \frac{1}{2}1 = \frac{5}{2}$. The cost of travel is 2, so the net expected value of the trip is $\frac{5}{2} - \frac{4}{2} = \frac{1}{2} > 0$. Hence, a risk-neutral person will go to the marketplace.

(c). Because x^2 exhibits increasing returns to money, this person is risk loving. Hence, the expected utility of spending \$2 and going to the market is now even greater than the expected utility of not going and saving \$2 in transportation costs, since there is the additional value from the risk. A risk-loving person would definitely go to market.

Note, that if you want to physically calculate the expected utility of going to the marketplace, you need a base wealth level to get an actual number. The preferences for money of $u(x) = x^2$, unfortunately are not well defined for negative numbers, so your answer will only make sense if you assume this person has at least a base wealth of 14.

46 Adverse selection. Specifically, although the average customer stays with VBS for 2+ years and generates profits of \$100 per year, the customer that your marketing promotion attracts may differ remarkably. The problem where the population distribution (of current or potential customers) differs from the actual distribution of customers who respond to your offer is known as adverse selection. In this case, it should not be surprising that some customers will join just to obtain the gift certificate, then they will quit the firm. Hence, their average tenure with the firm and their average contribution to the firm's profits is considerably smaller than the typical new customer who signs up without any promotion. Hence, adverse selection problems may render the marketing promotion unprofitable.

47 The expected utility of the sure-thing (i.e., no gamble) is

$$u(1,000,000) = \sqrt{1,000,000} = 1000 \text{utils}.$$

If she takes the gamble, her wealth is either $x_1 = 1,000,000 - 990,000 = 10,000$ or $x_2 = 1,000,000 + 2,240,000 = 3,240,000$. The expected utility of the gamble is therefore

$$E[u(\tilde{x})] = \frac{1}{2}\sqrt{10,000} + \frac{1}{2}\sqrt{3,240,000} = \frac{100}{2} + \frac{1800}{2} = 950.$$

Hence, the gamble offers too much risk (for the given return), so she should decline the gamble.

48 A one-line system will have the same expected time as a multi-line system. The variance of the one-line system, however, will be lower. Hence, if people have preferences for time which

are similar to their preferences for money, then risk-averse people (travelers with diminishing marginal utility of time) will prefer the one-line system while risk-loving people (increasing marginal utility of time) will prefer the multi-line system. This idea of lower risk is directly related to diversification as we combine several independent random variables (the time per traveler to get approval from a passport control agent).

49 True. First, note that the LR marginal cost curve must be below the LR average cost curve because it is decreasing (i.e., $MC_{LR}(q) < AC_{LR}(q)$), while the SR marginal cost curve must be above the SR average cost curve because the latter is increasing (i.e., $AC_{SR}(q) < MC_{SR}(q)$). Second, note that $AC_{LR}(q) \leq AC_{SR}(q)$ is always true because capital is flexible in the long-run. Hence, combining inequalities we have

$$MC_{LR}(q) < AC_{LR}(q) \leq AC_{SR}(q) < MC_{SR}(q).$$

50 Marginal cost is $2q$; average cost is $q + 16/q$. The minimum occurs at $q = 4$, where these two equations intersect.

51 FALSE (or at least uncertain). This answer would be true if there were no substitution possibilities. Since there usually are, it seems likely that costs will go up by less than \$25.

52 This is a case of a competitive, price-taking firm. (a) $MRTS = \frac{MPP_L}{MPP_K} = \frac{0.1}{0.3} = \frac{1}{3}$. The tangency condition for a cost-minimizing firm is

$$MRTS = \frac{w}{r},$$

and so an economically efficient firm facing wages and rental rates as given should have its $MRTS = \frac{150}{300} = \frac{1}{2}$. Since this is greater than $\frac{1}{3}$, the firm is employing too much labor. (b) In the short run only labor is variable, and so $MC_{sr} = \frac{w}{MPP_L} = \frac{150}{0.1} = \1500 .

53 False. There are two effects to consider. First, because capital has no economic cost in the short run, the short-run opportunity cost will be lower than long-run opportunity cost when the capital stock is at the long-run cost-minimizing level. On the other hand, in the short run the capital stock may not be at the cost-minimizing level, so the long-run cost may be lower given the more efficient level of capital. Hence, the relationship can go the other way.

54 True. The short-run marginal cost is simply $MC_{sr} = \frac{w}{MP_L(\bar{K}, L)}$, where L is such that $q = f(\bar{K}, L)$. Hence, as the wage increases, short run marginal cost goes up by the same amount. In the long run, however, the capital stock is also adjusted, so it is possible that the MP_L increases in greater proportion than the wage given this adjustment, which would cause the marginal cost to decline.

55 True. The formulae short-run and long-run marginal costs differ only in the chosen inputs, (K, L) , i.e.,

$$MC = \frac{w}{MP_L(K, L)} = \frac{r}{MP_K(K, L)}.$$

The choice of inputs is unaffected by a 10 percent increase in input prices since the relative input prices are constant. The numerator increases by 10 percent without any change in the denominator. Hence, marginal costs rise by 10 percent in both settings.

It is true that you may shut down production entirely due to the input price increases (i.e., your average costs may exceed price), but we can still say that at any given output, your marginal cost of production increases by 10 percent in both short-run and long-run.

56 True. Because the average cost curve is flat, it is neither rising nor falling. Hence, the marginal cost curve can be neither above it nor below it. Therefore, the marginal cost curve must lie coincident with the average cost curve: $MC(q) = AC(q)$. Hence, marginal cost is also flat.

57 Uncertain. If output less than doubles when *both* labor and capital inputs are doubled, then the technology exhibits decreasing returns to scale. Here, we are told that capital is held constant, so we cannot be certain. Indeed, if the marginal productivity of capital is high enough, doubling capital and labor together may cause output to more than double, implying that the technology exhibits increasing returns to scale.

58 False. The short-run marginal cost of output is given by the formula

$$MC_{sr}(q) = \frac{w}{MP_L}.$$

It will increase exactly 10 percent (because wages increase 10 percent). the change in the cost of other fixed inputs has no impact on short-run marginal cost because these inputs are sunk in the short-run.

59 \$10. By seeing the Clapton concert, you give up two things. First, you give up your Clapton ticket which you won. Second, you give up the consumer surplus you would have obtained had you attended the Dylan concert. Because the Clapton ticket is non-transferable, its value if you don't use it is zero. Thus, the opportunity cost of the ticket is zero. The consumer surplus you would have obtained had you gone to the Dylan concert is $CS = 50 - 40 = 10$. This is an opportunity cost. Add them together and you get the opportunity cost for attending the Clapton concert: $0 + 10 = 10$.

An aside: I learned of this question and the discussion that follows from Professor Robert Frank. Two researchers, Ferraro and Taylor, used a variant of this question to measure how well opportunity cost is taught and understood by economists. The question they used is simpler because it asked people to choose from four multiple-choice answers. They posed

their version of this question to different groups of people to see how well they understood opportunity cost. Surprisingly, the evidence indicates that most economists do not understand opportunity cost. Among professional economists (PhD's, Professors, etc.), only 21.6 percent could answer it correctly, which is less than what a random choice among the four choices would deliver. University students who have completed a course in economics do even worse; only 7.4 percent get it correct. We economists, and I am definitely including myself, need to do a better job teaching this idea. One of my goals is that by the end of this course everyone understands and can apply the concept of opportunity cost.

60 Uncertain. [This is a variation of an opportunity-cost question I asked on a previous final exam.] There are at least a few consistent ways to answer this question.

First, consider the net gain from each choice, ignoring the opportunity cost of time. If U_{CP} is your monetary value of going to the Coldplay concert, then the stand-alone consumer surplus from going to Cold Play for a price of \$50, ignoring your opportunity cost of time, is

$$CS_{CP} = U_{CP} - 50.$$

Similarly, if you decide to go to the U2 concert you must pay \$100 for a ticket, so your stand-alone consumer surplus from going to U2 for a price of \$100, ignoring the opportunity cost of time, is

$$CS_{U2} = U_{U2} - 100 > 0,$$

which we know is positive. Because you can only choose one of these activities, you should go to the U2 concert if $CS_{U2} > CS_{CP}$, which is equivalent to $U_{U2} - U_{CP} > 50$.

As a fact, we are told that absent the free ticket, you would have gone to the U2 concert. Hence,

$$U_{U2} - 100 > U_{CP} - 75.$$

The \$75 is here because that is what you would have to pay for the Coldplay ticket if it were not given to you for free. Thus, we can assume $U_{U2} - U_{CP} > 25$. Notice the difference $CS_{U2} - CS_{CP} = U_{U2} - U_{CP} - 50$. Whether or not this is positive or negative depends upon whether or not $U_{U2} - U_{CP} > 50$. But we cannot tell this from the question. From the facts of the question we only know that $U_{U2} - U_{CP} > 25$.

An alternative way to get the same answer is to think about just one choice, whether or not to attend the U2 concert, but carefully calculate the opportunity cost of U2 to include the loss of surplus from not attending the Coldplay concert. In short, the opportunity cost of attending the U2 concert is \$100 plus the stand-alone consumer surplus that you would have had going to the Coldplay concert. (You can't do both.) On net, you should go to the U2 concert if

$$U_{U2} - 100 - CS_{CP} > 0.$$

But using our formula for the stand-alone consumer surplus, CS_{CP} , this inequality is simply $U_{U2} - U_{CP} - 50 > 0$, the same as above.

We could also have considered the choice of going to the Coldplay concert in isolation. Here, we would want to subtract the \$50 that could be obtain from selling the ticket and also the

stand-alone consumer surplus of the U2 concert (an opportunity cost) which you must give up to go to the Coldplay concert. You will obtain the same inequality.

61 (a). The firm chooses an output such that $p = MC(q)$. Here, price is \$9 and marginal cost, a function of quantity, is $3 + 2q$. Equating price and marginal cost: $9 = 3 + 2q$. Solving for quantity, $q = 3$.

(b). Producer surplus is equal to the area below the market price (i.e., \$9) and above the average cost curve (i.e., $3 + q$). Therefore, producer surplus is equal to $(9 - (3 + q))(q) = (3)(3) = \9 .

62 True. At the optimal point of production, $MR = MC^O$, so MC must be above the average cost curve also. But when MC^O is above AC^O , the average cost curve is increasing.

63 The total marginal cost of producing an item in August for delivery in November is $q+2$; the total marginal cost of producing an item in September for delivery in November is $q+1$; and the total marginal cost of producing an item in October for delivery in November is q . Equating these marginal costs to the marginal revenue of 20 yields an optimal production of 18,000 toys in August, 19,000 in September, and 20,000 in October.

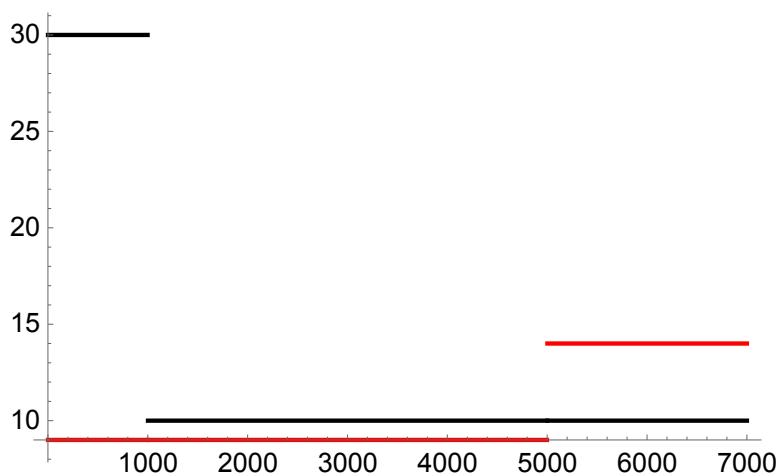
64 False. First consider the model of competition. In the long run, under competition, firms will produce where average costs are minimized. The long-run average cost curve is formed by determining the minimum cost at every level of output. In the short run, the firm might not be producing the optimal long-run output. So, if there are any fixed factors of production, the firm does not always produce where long-run average cost is minimized. In the case of monopoly, it is even easier to see why the statement is false. In such a setting, without entry, $MR=MC$, which tells us nothing about the relationship of output and the AC curve.

65 (a). The marginal cost of producing is \$9 per case, up to 5000 cases; it then rises to \$14 per case from 5001-7000 cases; it is infinite after 7000 cases per week (given your capacity). To write it as a function, we have

$$C^O(q) = \begin{cases} 0 & \text{if } q = 0 \\ 1000 + 9q & \text{if } 0 < q \leq 5000 \\ 1000 + 9(5000) + 14(q - 5000) = 14q - 24000 & \text{if } 5000 < q \leq 7000 \\ \infty & \text{if } q > 7000 \end{cases}$$

Given you receive \$10 per case, for any $q \leq 5000$, the marginal cost is 9 which is less than $MR = 10$, so you should produce those units. For any $q > 5000$, $MR = 10$ but MC increases to 14. Thus the marginal output rule says do not increase production beyond 5000. Hence, the profit maximizing level of output is 5000 cases per week, assuming you produce. At such a level of production, your weekly revenues are \$50,000 and your weekly costs are $\$1000 + 5000 \cdot 9 = \$46,000$. Profits are thus \$4000 per week.

(b). You should produce 1000 cases of brand aspirin and 4000 cases of generic aspirin. The marginal costs are the same as in (a) above. The key is to note that if you produce only five days a week (as hypothesized), your revenues are $\$30 \cdot 1000 + \$10 \cdot 4000 = \$70,000$ (since you would certainly sell one day's of output as branded aspirin). If you produce on Saturday as well, your revenue goes up by $\$10 \cdot 1000 = \$10,000$. Thus, the marginal revenue for the weekend production is only $\$10,000$, *not* $\$30,000$! Your marginal cost of an extra day of weekend production is still $\$14 \cdot 1000 = \$14,000 > MR = \$10,000$, so you should not produce on Saturday. Below is a graph indicating the marginal cost (in red) and the marginal revenue (black) curves. Notice that they cross at $q^* = 5000$.



66 Marginal cost is 10 for the first 30 units, then 15 for the next 20, and then 20 for all units above 50. Marginal revenue is found by allocating output to the high demand market first – market *B*. Thus, marginal revenue is 17 for the first 20 units, and 12 for the next 40 units. The first 20 units will be produced since marginal revenue exceeds marginal cost (i.e., $17 > 10$). The next 10 units produced will be allocated to market *A* with a price of $p_A = 12$ since this still exceeds marginal cost of 10. All additional units have marginal costs (15 or 20) which exceed marginal revenue (12). Hence, only 30 units will be produced: 10 will be sold in market *A* and 20 will be sold in market *B*.

67 (a). If the firm chooses to produce, it will choose q_{sr}^* to satisfy the marginal-output rule. $MR = p = 20$ and $MC_{sr}(q) = 4q$, so

$$20 = 4q_{sr}^*,$$

or $q_{sr}^* = 5$. At this level of output, however, short-run economic profit is

$$\pi_{sr}(q_{sr}^*) = 20(5) - (60 + 2(5)^2) = 100 - 110 = -10 < 0.$$

Hence, the firm chooses not to produce in the short-run, given its current capital stock. The firm *shuts down*.

(b). In the long-run, if the firm decides to produce, it will produce at q_{lr}^* to satisfy the marginal output rule:

$$20 = 2q_{lr}^*,$$

or $q_{lr}^* = 10$. At this level of output, the firm's long-run economic profits are

$$\pi_{lr}(q_{lr}^*) = 20(10) - (90 + (10)^2) = 10 > 0,$$

so our firm will decide to produce $q_{lr}^* = 10$ in the long-run.

68 False. Profit can be increased by *reducing* output because the marginal revenue, p , from production is less than the marginal cost. (This is the marginal output rule.) Because $p > AC$, we know that the firm should not shutdown, so the marginal output rule should be used and the firm should produce. (This is an application of the shutdown rule)

69 TRUE. As an example, the price may exceed average variable costs $p > AVC(q^*)$, but not long-run average costs because the latter includes the cost of capital, $p < AC_{Lr}$.

70 (a). The minimum average cost for each firm is 0, so the firm will be willing to supply at any positive price at the output for which $p = C'_{sr}(q)$. Thus, $p = 4q$ characterizes each firm's supply curve.

(b). To determine the market supply curve, we add up each of the 600 firms' individual supply functions. Because $q^s = \frac{p}{4}$ for each firm, market supply is

$$Q^s = 600\frac{p}{4} = 150p.$$

Equating supply and demand, we have

$$1200 - 50p = 150p,$$

so the short-run equilibrium price is $p^* = 6$. The short-run equilibrium quantity is therefore $Q^* = 1200 - 50(6) = 900$.

(c). Because this is a constant-cost industry, we know the long-run market price will equal long-run minimum average cost. This occurs where $MC_{lr}(q) = AC_{lr}(q)$. Specifically,

$$4q + 1 = 2q + \frac{2}{q} + 1,$$

so the minimum average cost occurs at $q = 1$. At this output, each firm's minimum average cost is $AC_{lr}^{min} = 5$. Thus, the equilibrium price is $p^* = 5$. At this price, equilibrium output is $Q^* = 950$. Given each firm is producing at $\hat{q} = 1$, this implies a total of 950 firms will produce.

71 This is a constant-cost industry, so we first find the equilibrium price p^* by calculating the minimum average cost of each firm. The min AC_{lr} is found by equating $MC_{lr}(q_{\min}) = AC_{lr}(q_{\min})$ which implies

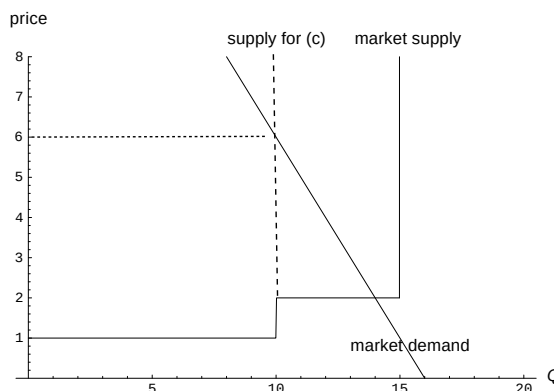
$$2q_{\min} = q_{\min} + \frac{9}{q_{\min}},$$

or $q_{\min} = 3$. At this output, $AC_{lr}(3) = 6$. Thus, the market price will be $p^* = 6$ given that there is a sufficiently large number of firms. Lastly, we need to compute how many firms (each producing 3) can supply cheese while maintaining a market price of $p^* = 6$. Specifically,

$$3 * n = 120 - 6$$

so $n^* = 38$.

72 (a). Verbally, it is a horizontal line at 1 which extends to the right to 10 MMH, then it jumps to 2 and extends to the right to 15 MMH, then it rises vertically. Graphically, it is the supply curve given in the solid line.



(b). The horizontal part of the second tier of the supply curve intersects the demand curve at $p = 2$. At this price, $2 = 16 - Q$, so $Q = 14$. Hence, 10 units are supplied by the efficient firms and 4 units are supplied by the remaining inefficient firms. See above graph.

(c). With only the efficient firms, the supply curve is horizontal at $p = 1$ up to $Q = 10$, then it is vertical. It is given by the thick, vertical dotted line in the graph. It intersects the demand curve at $Q = 10$ and a price of $p = 16 - 10 = 6$.

73 (a). The marginal cost schedules are as follows:

Quantity	Firm A	Firm B	Firm C
1	1	1	1
2	2	3	1
3	3	4	2
4	4	5	3
5	5	6	4
6	6	7	5
7	7	8	6

Thus the supply schedules are as follows:

Price	Firm A	Firm B	Firm C	Total
1	1	1	2	4
2	2	1	3	6
3	3	2	4	9
4	4	3	5	12
5	5	4	6	15
6	6	5	7	18
7	7	6	7	20
8	7	7	7	21

(b). At a price of 5, supply and demand are equated at 15.

(c). Firms 1, 2, and 3 produce 5, 4, and 6 respectively.

74 (a). We find the equilibrium price and quantity by equating the short-run quantity supplied and demanded:

$$96 - p = 3p,$$

$$4p = 96,$$

$$p^* = 24.$$

Hence, $Q^* = 96 - 24 = 72$.

(b). First part. In a competitive market, each firm produces in the short-run where $p^* = MC_{sr}(q^*)$, providing that $p^* \geq AC(q^*)$. Using the first expression and the price $p^* = 24$ calculated in part (a), we have,

$$24 = 8q^*,$$

so $q^* = 3$ units per firm. Now, checking the shutdown rule, at this level of production the average opportunity cost (i.e., AVC) is $AC_{sr}^O(3) = 4(3) = 12$ which is less than $p^* = 24$. Therefore, each firm in the industry produces $q^* = 3$.

Second part. Given that there is a total of 72 units produced in the industry and each firm produces 3, there are $N^* = \frac{72}{3} = 24$ firms in the industry.

(c). There are at least 2 ways to calculate this. The most direct is to note that each firm's short-run economic profit is

$$\pi_{sr} = (24)(3) - 4(3)^2 = 36.$$

There are 24 firms, so the total producer surplus is $(24)(36) = 864$.

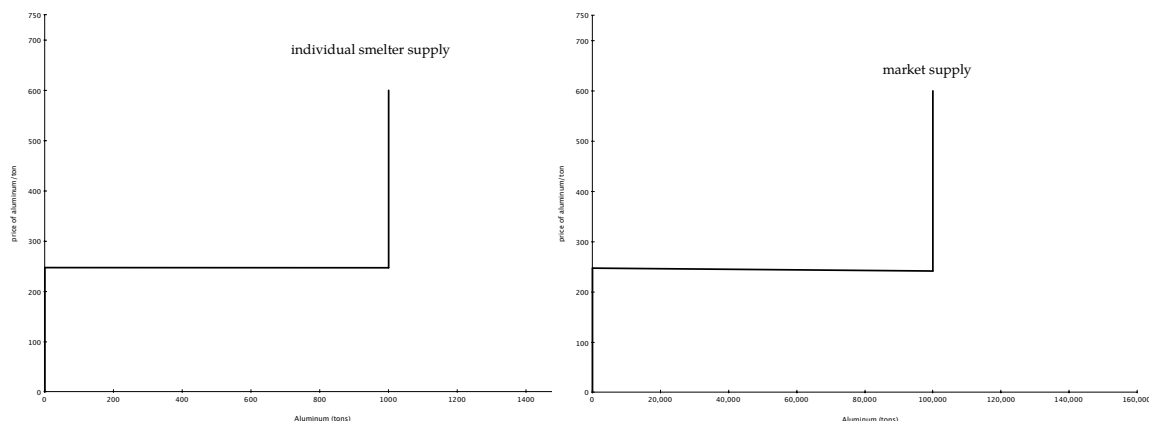
Alternatively, you could take the market supply curve, $Q^s = 3p$, and then find the area above the curve and below $p^* = 24$. (As an aside, each firm's supply curve is derived from the marginal output rule, $p = 8q$, so $q^s = p/8$. Adding up 24 firms, we have $Q^s = 24\frac{p}{8} = 3p$ as given above.) In this case, we have a triangle of height 24 and width 72. The area is $\frac{1}{2}(24)(72) = 864$.

(d). The demand elasticity at the equilibrium point is

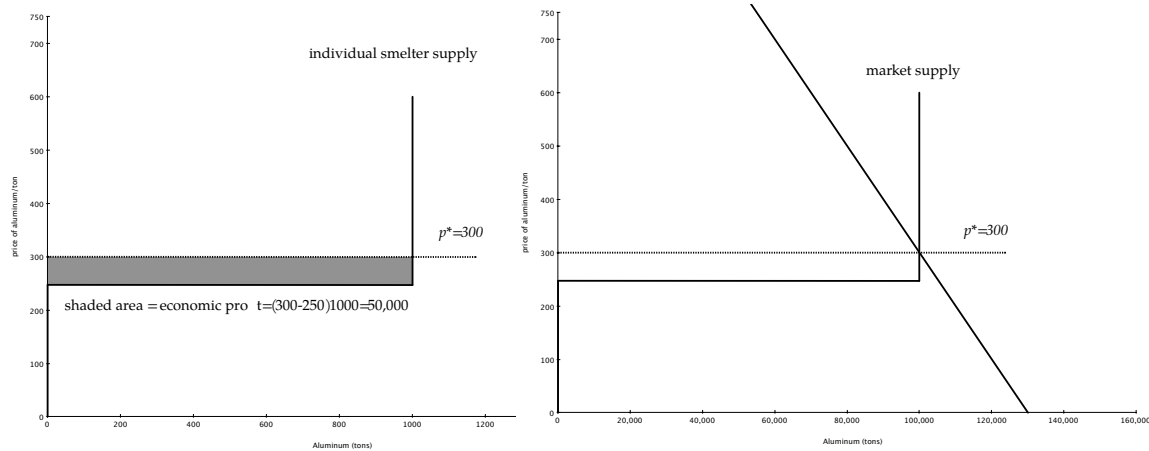
$$\varepsilon_d = \frac{dQ^d}{dp} \frac{p^*}{Q^*} = (-1) \frac{24}{72} = -\frac{1}{3}.$$

(e). From (d), demand is inelastic. A leftward shift of the supply curve would raise the market price by a greater percentage than output would fall, and hence inelastic demand implies an increase in market revenues.

75 (a). Each firm is willing to produce at capacity at a critical price of \$250 per ton. Thus, supply has a simple backwards L-shape. The industry supply is just the sum of 100 individual smelter supply curves. In the two figures below, note that the scale of the horizontal axes differ by a factor of 100.



(b). Assuming that the price is greater than \$250, we know that the market will supply $Q^s = 100(1000) = 100,000$. For demand to equal this output, the equilibrium price must be $p^* = 300$.



Short-run economic profit at this price is

$$\pi_{sr} = (p^* - AC_{sr}^O)q^* = (300 - 250)(1000) = 50,000.$$

(c). Given the price will still support all 100 smelters and total output will equal 100,000 tons, we need only find the price for which demand will equal 100,000 tons. Hence

$$p^* = 1500 - 0.01(100,000) = 500.$$

At this price, each firm's short-run economic profit becomes

$$\pi_{sr} = (500 - 250)(1000) = 250,000.$$

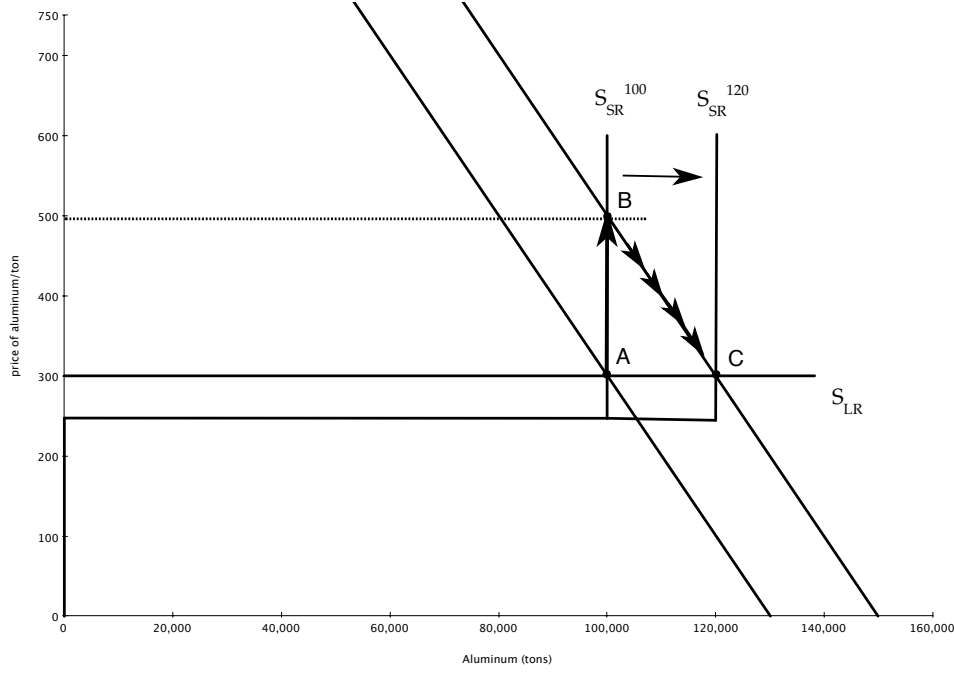
(d). Because this is a constant-cost industry, the equilibrium price will equal the minimum long-run average opportunity cost. To find this number, notice that the long-run average cost function is decreasing for all $q \leq 1000$, so it is at a minimum for $\hat{q} = 1000$. The value of the long-run average opportunity cost at capacity is

$$\min AC_{lr} = \frac{250(1000) + 50,000}{1000} = 250 + 50 = 300.$$

At $p = 300$, long-run profits exactly equal zero. It follows that any firm that expects the price of aluminum to be greater than 300 would want to invest in a smelter. Hence, the long-run supply for the industry is a horizontal line at 300.

(e). At the initial demand function, the equilibrium price is $p^* = 300$, which leaves no long run profit from operating a smelter. Hence, supply is 100 smelters, each producing 1000 tons. Here, all firms are in short-run and long-run equilibrium. (This is point *A* in the figure below.) When demand unexpectedly shifts outward, the existing 100 smelters all make increased profit as the price rises to $p = 500$. (This is point *B*.) Now we are in short-run equilibrium, but not long-run equilibrium: At this price, firms will find it profitable to build

additional smelters. As smelters are built and capacity is added, however, the short-run supply curve shifts outward and the price decreases (the movement from point B to C in the figure). Smelters continue to be built until the price decreases back to $p = 300$. (Point C .) We now have 120 smelters operating at capacity and we are again back in long-run (as well as short-run) equilibrium.



76 (a). First, we compute each firm's minimum average cost. At the output which achieves this minimum, we have $MC_{sr}(\hat{q}) = AC_{sr}^O(\hat{q})$, and thus

$$8\hat{q} + 2 = \frac{4\hat{q}^2 + 2\hat{q} + 4}{\hat{q}},$$

so $\hat{q} = 1$. At this output, a firm's minimum average cost is simple $AC_{sr}^O(1) = 10$. We may thus conclude that for any price $p < 10$, no firm will supply any output, but for a price $p > 10$, each firm will supply.

To determine the amount each firm supplies, we use the marginal output rule: $MC_{sr}(q) = p$. In our present case, we have

$$8q^s + 2 = p,$$

so $q^s = \frac{p-2}{8}$, providing that $p > 10$. Altogether, we can write the firm's supply function as

$$q^s = \begin{cases} 0 & p < 10 \\ \frac{p-2}{8} & p > 10. \end{cases}$$

(b). If there are 800 firms in the short run, we simply multiple the supply function for each firm by 800. We immediately obtain

$$Q^s = \begin{cases} 0 & p < 10 \\ 100p - 200 & p > 10. \end{cases}$$

Given the demand function, equilibrium requires that $Q^d = Q^s$, so

$$2200 - 100p = 100p - 200,$$

assuming that the p which satisfies this equation is indeed greater than 10. Solving for p^* , we obtain

$$2400 = 200p,$$

so $p^* = 12$. At this price, each firm produces $q^s = (12 - 2)/8 = 1.25$ units, and the market supply and demand is equal to $Q^* = 1000$.

(c). Because this is a constant-cost industry, we know the long-run market price will equal long-run minimum average cost. This occurs where $MC_{lr}(q) = AC_{lr}^O(q)$. Specifically,

$$4\hat{q} = \frac{2\hat{q}^2 + 8}{\hat{q}},$$

so the minimum average cost occurs at $\hat{q} = 2$. At this output, each firm's minimum average cost is $AC_{lr}^O min = AC_{lr}^O(2) = 8$. Thus, the equilibrium price is $p^* = 8$. At this price, equilibrium output is $Q^* = 2200 - 100(8) = 1400$. Given each firm is producing at $\hat{q} = 2$, this implies a total of 700 firms will produce.

77 (a). In the high demand state (assuming that industry capacity is exhausted and each of the 100 firms supplies 1 unit of output), the price is

$$p_h = 310 - (100)1 = 210.$$

In the low demand state, if every firm supplied its capacity, the market price would be $p_l = 110 - 2(100)1 = -90 < 0$, so clearly not all firms will supply at full capacity. Instead, price will be driven down to short-run marginal cost, $MC_{sr} = 10$. Hence, $p_l = 10$. At this price, aggregate supply must equal 50. (Aside: There are many ways that the 50 units could be produced in equilibrium (e.g., each firm produces half a unit, or 50 firms produce at capacity and 50 firms produce zero.) Regardless of how the 50 units are allocated, each firm makes zero short-run economic profit.

(b). First, calculate the expected profit of a firm in this industry. In the low-demand state, a firm's short-run economic profit (ignoring capital costs, of course) is $\pi_l = 0$, because $p_l = 10 = AC_{sr}^O$. In the high demand state, $p_h = 210$, so profit is $\pi_h = (p_h - AC_{sr}^O)q = (210 - 10)1 = 200$. Expected short-run profit is $\pi^{sr} = \frac{1}{2}(0) + \frac{1}{2}(200) = 100$. Expected profit exceeds the per-period opportunity cost of capital, 75

(c). Firms will continue to enter until the long-run average (opportunity) cost of production (this includes the per-period cost of capital, \$75) equals the long-run expected price. Alternatively, this firms will continue to enter until the long-run economic profit of the industry is zero.

Using average opportunity costs, we have $AC_{lr}^O = 75 + 10 = 85$. Long-run expected price is

$$E[p] = \frac{1}{2}(10) + \frac{1}{2}(p_h) = \frac{1}{2}(10) + \frac{1}{2}(310 - N) = 160 - \frac{N}{2}.$$

Equating, we have

$$85 = 160 - \frac{N}{2},$$

or $N^* = 150$.

Alternatively, using long-run economic profits, we know that profits are zero in the low-demand state. Thus, long-run economic profit is

$$E[\pi_{lr}] = \frac{1}{2}(0) + \frac{1}{2}\pi_{sr}^h - 75 = \frac{1}{2}(310 - N - 10) - 75 = 75 - \frac{N}{2}.$$

Thus, $N^* = 150$.

78 (a). $AC=MC$ at the level of production at which AC is minimized. Because short-run AVC is minimized at $q = 9$ and $MC = 40 - 12(9) + 9^2 = 13$ at that point, $AC_{sr}^O = 13$. Because long-run AC is minimized at $q = 12$ and $MC = 40 - 12(12) + 12^2 = 40$ at that point, $AC_{lr} = 40$.

(b) With a price of 40, total output must be $Q = 160 - p = 120$. Thus, each firm produces $q = 12$ in the long-run with $AC=40=p$, and so $\pi = 0$. This is an equilibrium. No outside firm wishes to enter; none wishes to leave.

(c)-(d). With foreign competition, $p = 20$. In the short-run, firms can produce at $AC = 13 < 20$, and so they stay in. In the long-run, these firms will shut down operations because they cannot cover their long-run average costs of 40.

79 (a). Each plant has a marginal cost of $2q$, so its supply curve is $q = P/2$. Since there are 10 plants/firms, the industry supply curve is $Q = 5P$. Equating supply and demand, the price is $P = 16$. Thus each firm produces 8 widgets.

(b). Average cost is $16/q + q$, and that is minimized when $q = 4$. The LRAC is then 8, so that price will be 8.

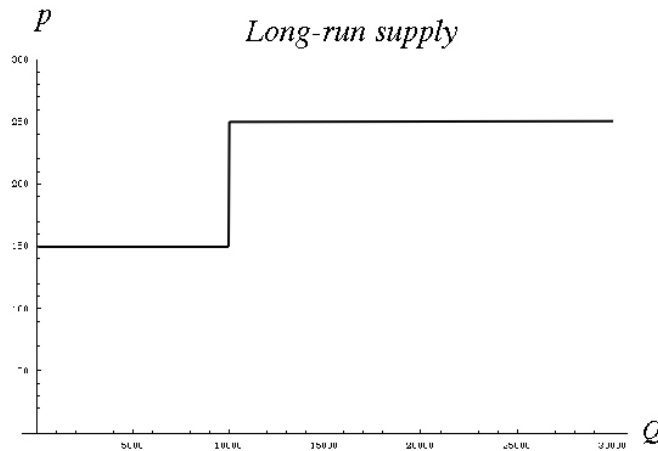
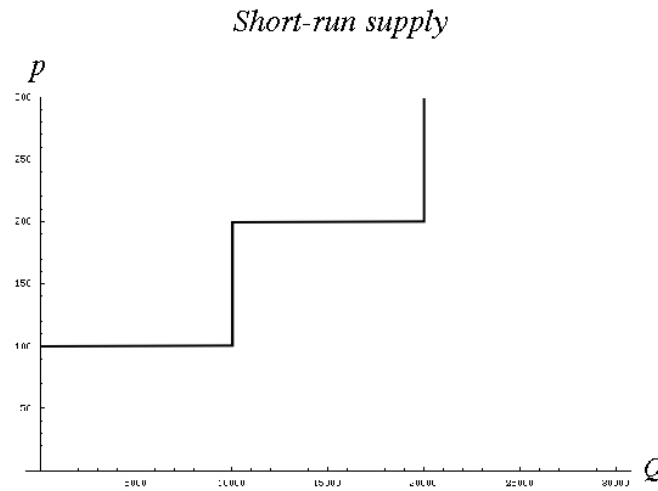
(c). 4.

(d). Total demand will be 160, so there will be 40 firms.

80 False. Firms will follow the marginal output rule and the shutdown rule. The marginal output rule in general will not have the firm choosing output at the point where short-run average opportunity cost is at a minimum. Rather, the firm will equate marginal cost to the price, which may or may not exceed the minimum average cost. If the market price is above or below the minimum average cost of the firm, the firm will not produce at this minimum.

81 (a). In the short-run, the 10 efficient firms are willing to supply all of their capacities (10,000 in total) at any price equal to or above 100 (short-run opportunity AC and MC). The 10 inefficient firms are willing to supply all of their capacities (another 10,000) at any price above 200 (their higher short-run opportunity AC and MC). Combining these outputs yields the short-run supply curve: 0 below $p = 100$, a horizontal segment of length 10000 at $p = 100$, and a horizontal segment of length 10000 at $p = 200$. Supply is vertical for any price once $Q = 20000$ (the combined capacities of all existing firms). [See figure, first panel.]

In the long-run, 10 efficient firms are each willing to produce the capacity of 1000 for at least $p = 150$ (determined by $AC_{lr}(1000) = 100 + \frac{50000}{1000} = 150$); an unlimited number of inefficient firms are willing to supply 1000 each for any price at least $p = 250$ (determined by $AC_{lr}(1000) = 200 + \frac{50000}{1000} = 250$). thus, the long-run supply curve is 0 for any price below 150, a horizontal segment of length 10,000 at price $p = 150$, and another infinitely long horizontal segment at $p = 250$. [See figure, second panel.]



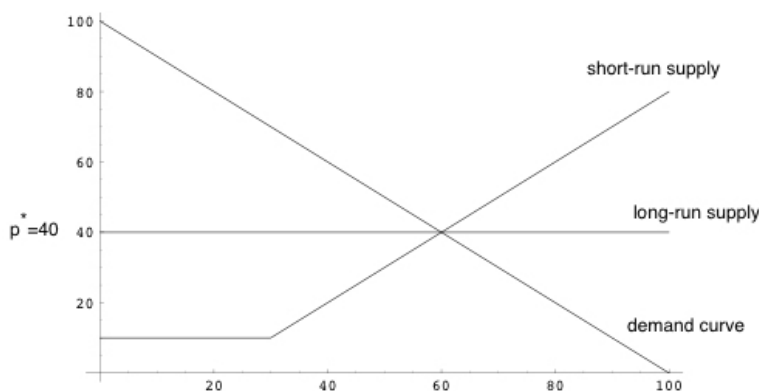
(b). Because both firms supply at the equilibrium price, the equilibrium market price must equal the minimum average cost of the inefficient firms. If it were higher, more inefficient firms would enter in the long run to drive the price downward; if the price is less than the minimum average cost of the inefficient firms, then some inefficient firms would exit the industry. Because $AC_{lr}(1000) = 200 + \frac{50,000}{1000} = 250$ is the long-run minimum cost for the inefficient firms, $p^* = 250$.

(c). (1). Initially, prices will rise above 250 to equate short-run supply with demand. Because the demand curve will shift backward next period, the long-run price will return to $p = 250$. (2). Next period, no additional entry or exit takes place because the demand curve is expected to return to its previous position. (3). In the short-run, all 20 firms will produce at capacity (hence, 20,000 in output is made). (4). All 20 firms will make short-run economic profits because price exceeds average variable cost for all of the firms (which is at most 200).

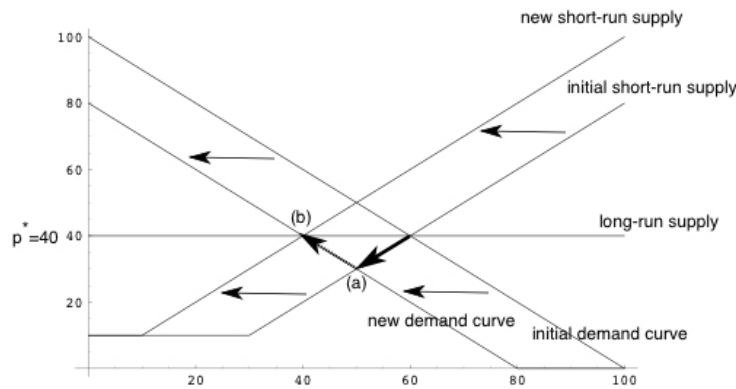
82 (a). Below is a constant-cost industry example in which each firm has long-run average

costs equal to 40 and minimum average variable costs equal to 10. Regardless of how you draw the three curves, your figure should have (i) all 3 curves intersect at the long-run equilibrium price, (ii) the short-run supply curve must be steeper than the long-run supply curve at the equilibrium point.

(Additionally, if the minimum average variable cost is positive, the short-run curve should intersect the vertical axis at a point strictly above zero. I did not deduct points if you were not specific about the behavior of the supply function at the vertical axis.)



(b). Now we shift the demand curve inward. The immediate change in price is to jump downward along the initial short-run supply curve to the intersection between the initial short-run supply curve and the new demand curve. This is labeled point (a) in the figure and the movement is indicated by the solid arrow line along the initial short-run supply curve. In the long-run, firms leave the industry because this temporary price is below the long-run average cost. Firms continue to leave the industry until the new short-run supply curve has shifted inward and intersects the point at which the new demand and long-run supply cross. This is point (b). This inward shift of the short-run supply curve implies that the price in the market will rise from the temporary reduction at point (a) to the final point at (b) where $p^* = 40$. (Note, that if you assumed an increasing-cost (or decreasing-cost) industry, the new long-run equilibrium price will be lower (higher, resp.) than the original long-run equilibrium price.)



83 (a). Each firm's marginal cost is $6q$, so it supplies $q = p/6$ for $p \geq 18$. The industry supply function is then $Q = (40/6)p$.

(b). (1). Equating supply and demand, we have $(40/6)p = 600 - 10p$, so the equilibrium price is 36.

(2). As seen above, each firm will produce 6 widgets.

(3). Each firm will have a profit of $(36)(6) - (27 + 3 \cdot 6^2) = 81$.

(c). The supply curve is a horizontal line at $p = AC'_{lr}(3) = 18$.

(d). (1). The price is 18.

(2). Each firm will be producing 3 widgets.

(3). Total market quantity is $Q = 600 - 10(18) = 420$. Thus, there will be $420/3=140$ firms.

(4). No economic profit will be made as $p = AC^o_{lr}(3)$.

84 TRUE/Uncertain. The statement is true if the industry is a constant-cost industry and so the supply curve is horizontal. But, for example, if it is an increasing cost industry, the supply curve is upward sloping, and so an increase in demand leads to an increase in price.

85 False. Firms will follow the marginal output rule and the shutdown rule. The marginal output rule in general will not have the firm choosing output at the point where short-run average opportunity cost is at a minimum. Rather, the firm will equate marginal cost to the price, which may or may not exceed the minimum average cost. If the market price is above or below the minimum average cost of the firm, the firm will not produce at this minimum.